

AI-based medical image tampering detection for secure IoMT systems

Project Report Submitted in Partial Fulfillment Of
The Requirements for The Degree of

Bachelor of Technology
In
Electronics and Communication Engineering
Of

Maulana Abul Kalam Azad University of Technology,
West Bengal

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2022-2026

Abstract

This report presents Project Part-II, an advanced and direct continuation of Project Part-I, “CNN-Based and Advanced LSB Steganography for Secure Medical Image Communication in IoMT Systems.” While Part-I established a robust framework for medical image confidentiality using CNN-based steganography, Advanced Least Significant Bit (LSB) embedding, and AES encryption, it left an important dimension of security unaddressed: the ability to verify whether a medical image received over an Internet of Medical Things (IoMT) network is authentic, unaltered, and free of malicious manipulation. Part-II closes this gap by proposing an AI-based medical image tampering detection framework that operates as the receiver-side counterpart to the sender-side steganographic pipeline developed previously.

The proposed system employs a hybrid architecture combining Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to classify medical images as authentic or tampered, to localize manipulated regions, and to detect sophisticated AI-generated deepfake medical images such as synthetically inserted or removed tumors. Because clinical adoption of automated security tools depends on transparency, the framework integrates Gradient-weighted Class Activation Mapping (Grad-CAM) to generate visual heatmaps that explain the model's decisions to radiologists and clinicians. The report reviews more than forty recent research contributions spanning medical image forensics, deepfake detection, Vision Transformers in medical imaging, explainable AI, medical image watermarking and authentication, and IoMT security, situating the proposed work within this broader research landscape.

The methodology chapter details the end-to-end pipeline, from dataset acquisition and preprocessing through CNN/ViT-based feature extraction, tamper classification, region localization, and Grad-CAM-based explanation generation, supported by mathematical formulations of the loss functions and evaluation metrics used. Implementation details covering the Python, TensorFlow, PyTorch, and OpenCV toolchain, along with the hyperparameters and training configuration, are presented alongside a discussion of expected experimental outcomes measured through accuracy, precision, recall, F1-score, ROC-AUC, and confusion-matrix analysis. The report further discusses real-world applications in telemedicine, digital pathology, medical insurance fraud prevention, and secure medical archiving, alongside a candid assessment of the framework's limitations and the future scope for federated learning, blockchain-based provenance, and edge-AI deployment.

Taken together, Part-I and Part-II form a complete, end-to-end secure IoMT pipeline: Part-I guarantees that patient information remains confidential during transmission, while Part-II guarantees that the received image can be trusted as authentic and clinically reliable. This two-phase, complementary design reflects the dual requirement of modern healthcare cybersecurity — confidentiality and integrity — and demonstrates a coherent, cumulative body of research suitable for deployment in next-generation, AI-secured medical imaging systems.

Keywords: Medical Image Tampering Detection, IoMT Security, Convolutional Neural Networks, Vision Transformer, Explainable AI, Grad-CAM, Medical Deepfake Detection, Image Authentication, AES Encryption, Steganography, Healthcare Cybersecurity.

Acknowledgement

We express our sincere gratitude to our project guide and the faculty of the Department of Engineering for their continuous guidance, encouragement, and constructive feedback throughout both phases of this research. Their mentorship during Project Part-I laid the foundation for the confidentiality-focused steganographic framework, and their continued support during Part-II has been instrumental in extending that foundation toward a complete, integrity-aware secure IoMT pipeline.

We also thank the Head of the Department and the college administration for providing the computational resources, laboratory access, and academic environment necessary to carry out this extended research. We are grateful to our peers for their valuable discussions during the literature review and implementation phases, and to our families for their patience and encouragement throughout the duration of this two-semester project.

Finally, we acknowledge the authors of the numerous research papers reviewed in this report, whose published work in medical image forensics, explainable AI, and IoMT security has directly informed the design decisions made in Project Part-II.

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List of Abbreviations

Abbreviation	Full Form
AES	Advanced Encryption Standard
AI	Artificial Intelligence
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CT	Computed Tomography
DICOM	Digital Imaging and Communications in Medicine
FL	Federated Learning
GAN	Generative Adversarial Network
Grad-CAM	Gradient-weighted Class Activation Mapping
IoMT	Internet of Medical Things
IoT	Internet of Things
LSB	Least Significant Bit
MRI	Magnetic Resonance Imaging
MSE	Mean Squared Error
PSNR	Peak Signal-to-Noise Ratio
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
ROI	Region of Interest
SSIM	Structural Similarity Index Measure
TinyML	Tiny Machine Learning
ViT	Vision Transformer
XAI	Explainable Artificial Intelligence

Chapter 1: Introduction

1.1 The Evolution of Digital Healthcare

Healthcare delivery has undergone a profound digital transformation over the past decade. Paper-based patient records have given way to Electronic Health Records (EHRs), film-based radiographs have been replaced by digital imaging modalities such as Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and digital X-ray, and diagnostic workflows increasingly depend on cloud storage, teleradiology, and remote consultation. This shift has improved accessibility, reduced diagnostic turnaround time, and enabled collaborative, multi-institutional care. However, it has also transformed the medical image from a static, physical artifact into a digital asset that travels across networks, is stored on shared servers, and is processed by automated pipelines — each of which introduces a potential point of compromise. Project Part-I of this research addressed one dimension of this transformation, namely the secure transmission of medical images using steganographic embedding; Project Part-II, documented in this report, addresses a second and equally critical dimension: verifying that the image received at the end of that journey is exactly the image that was sent.

1.2 The Internet of Medical Things (IoMT)

The Internet of Medical Things refers to the interconnected ecosystem of medical devices, imaging equipment, hospital information systems, and cloud platforms that communicate over standard network protocols to support modern healthcare delivery. IoMT encompasses everything from bedside patient monitors and wearable sensors to full Picture Archiving and Communication Systems (PACS) that store and transmit medical images across hospital networks and, increasingly, across international borders for teleradiology and second-opinion services. The convenience and clinical benefit of this connectivity come at a cost: every additional network hop, every cloud storage bucket, and every interoperability gateway is a surface across which an adversary may intercept, delay, or alter data. IoMT security research therefore treats confidentiality, integrity, and availability as three co-equal pillars, and this report is chiefly concerned with the second of these — integrity — as it applies specifically to diagnostic images.

1.3 Medical Image Security: Confidentiality and Integrity

Medical image security is commonly decomposed into two complementary objectives. Confidentiality ensures that sensitive patient information embedded within, or associated with, a medical image cannot be read by unauthorized parties; this was the central concern of Project

Part-I, which used CNN-based steganography, Advanced LSB embedding, and AES encryption to hide patient identifiers and diagnostic notes inside stego-images while preserving diagnostic image quality. Integrity, in contrast, ensures that the image content itself has not been altered — whether by transmission error, deliberate forgery, or malicious substitution — between the point of acquisition and the point of clinical review. A system can achieve perfect confidentiality while still failing catastrophically on integrity: an attacker who cannot read the hidden patient ID may nevertheless succeed in subtly editing the visible anatomy of the scan, for instance by inserting or erasing a lesion, without ever needing to break the encryption layer. This observation is the central motivation for Project Part-II.

1.4 Cybersecurity in Hospitals

Healthcare has become one of the most heavily targeted sectors in global cybersecurity incident reporting. Hospitals operate large, heterogeneous device fleets, often running legacy software that cannot be patched without risking clinical downtime, and they hold high-value data that is attractive to ransomware operators and, increasingly, to actors interested in insurance fraud or clinical sabotage. Reported cyber incidents affecting the healthcare sector have shown a marked year-on-year increase, and a substantial share of connected healthcare devices have been found to carry exploitable network vulnerabilities. Within this threat landscape, medical images occupy a uniquely sensitive position: unlike a stolen record that merely exposes private information, a tampered image can directly cause a wrong diagnosis, a missed intervention, or an unnecessary procedure, translating a purely digital attack into direct physical harm to a patient.

1.5 Medical Image Attacks: A Taxonomy

Attacks against medical images can be broadly grouped into four categories relevant to this project. First, passive interception attacks, in which an adversary reads confidential data without modifying it — the threat Part-I was designed to defeat through encryption and steganographic concealment. Second, active tampering attacks, in which pixel-level content is deliberately altered, such as splicing, copy-move duplication, or region removal/insertion, to change the apparent diagnosis. Third, synthetic generation attacks, in which an entirely fabricated medical image is produced using generative models (GANs or diffusion models) and injected into a workflow to defraud an insurer or fabricate a clinical record; the CT-GAN study by Mirsky et al. demonstrated that convincingly falsified tumors could be injected into or removed from real CT scans well enough to deceive expert radiologists and automated

diagnostic tools alike. Fourth, provenance attacks, in which an image is genuine but its claimed source, patient association, or acquisition timestamp is falsified. Project Part-II is designed primarily to counter the second and third categories, while remaining compatible with the confidentiality guarantees Part-I already provides against the first.

1.6 Confidentiality versus Integrity: Why Tampering Detection Matters

It is tempting to assume that once a communication channel is encrypted and the payload is steganographically concealed, the security problem is solved. This assumption does not hold for medical images, because encryption and steganography are designed to protect data in transit, not to certify the visual truthfulness of the image after decryption. A stego-image can decrypt perfectly, and the hidden patient record can be extracted flawlessly, while the underlying anatomical content has nonetheless been manipulated at the source or at an intermediate node before encryption ever occurred, or via a compromised endpoint after decryption. Because a physician's diagnostic decision depends entirely on the pixel content of the image rather than on the surrounding metadata, an integrity failure at this level is clinically indistinguishable from a genuine finding unless an independent detection mechanism is in place. This is precisely the gap that AI-based tampering detection is designed to close.

1.7 Existing Challenges

Several practical challenges complicate the design of a medical image tampering detection system. Medical images differ substantially from natural photographs: they are often single-channel or pseudo-colour, have high dynamic range, and contain diagnostically meaningful fine structures (such as micro-calcifications or small lesions) that general-purpose forgery detectors trained on natural images are not tuned to preserve. Publicly available tampered-medical-image datasets remain comparatively scarce next to natural-image forgery benchmarks such as CASIA or Columbia, forcing researchers to generate synthetic tampering themselves. Explainability is a further challenge: a black-box classifier that merely outputs “tampered” or “authentic” is of limited clinical value, because a radiologist must be able to see which region triggered the alert before acting on it. Finally, tampered regions in medical images are frequently small relative to the overall image, placing the detection task in a harder “small-object” regime where popular explainability techniques such as Grad-CAM are known to be less reliable, a finding this report examines in detail in Chapter 3.

1.8 Motivation for the Present Research

The motivation for Project Part-II follows directly from the limitations identified at the conclusion of Project Part-I. Having built a system capable of hiding confidential data inside medical images with high fidelity (as measured by PSNR, SSIM, and MSE), the natural and necessary next step is to build the complementary system that verifies the received image has not been tampered with. Modern telemedicine platforms, cloud-based PACS systems, and cross-institutional data-sharing agreements all increase the number of intermediate points at which an image could, in principle, be altered before it reaches the reviewing clinician. An AI-based, explainable tampering detection layer therefore serves as a necessary trust anchor for any IoMT deployment that aspires to end-to-end security rather than confidentiality alone.

1.9 Research Objectives

The specific objectives of Project Part-II are as follows:

- To design and evaluate a CNN and Vision Transformer-based framework capable of classifying medical images as authentic or tampered.
- To localize the specific tampered regions within a manipulated medical image, rather than providing only an image-level authenticity label.
- To integrate Grad-CAM-based explainability so that clinicians can visually verify the model's reasoning before trusting an automated flag.
- To extend detection coverage to AI-generated deepfake medical images, in addition to conventional pixel-level tampering.
- To integrate the proposed detection framework with the steganographic and encryption pipeline developed in Project Part-I, forming a single, coherent, end-to-end secure IoMT architecture.
- To evaluate the proposed framework using standard classification metrics (accuracy, precision, recall, F1-score, ROC-AUC) and to compare its performance against baseline methods reported in the literature.

1.10 Scope of the Project

The scope of Project Part-II is limited to the design, implementation, and experimental evaluation of an AI-based tampering detection and authentication framework for medical images (CT, MRI, and X-ray) operating on the receiver side of the IoMT pipeline established in Part-I. The project covers dataset preparation and augmentation, CNN and Vision Transformer model design, Grad-CAM-based explainability, and evaluation against synthetic

and, where available, publicly documented tampering scenarios. The scope explicitly excludes redesigning the steganographic embedding or AES encryption mechanisms of Part-I, which are treated as a stable, already-validated foundation; it also excludes full-scale clinical trial validation, hospital information system integration, and regulatory certification, all of which are identified in Chapter 10 as future work beyond the present academic project.

1.11 Problem Statement

Existing medical image steganography and encryption methods, including the framework developed in Project Part-I, successfully protect the confidentiality of patient data embedded within medical images but provide no mechanism to verify the integrity and authenticity of the image content itself after transmission. There is consequently a need for an intelligent, explainable, AI-based framework capable of detecting pixel-level tampering and AI-generated deepfake manipulation in medical images, localizing the affected regions, and presenting this information to clinicians in a transparent, trustworthy manner, so that the overall IoMT pipeline achieves both confidentiality and integrity.

1.12 Research Methodology

The research methodology adopted for Project Part-II follows a structured, five-stage process. The first stage consists of an extensive literature review covering medical image forensics, medical deepfake detection, Vision Transformers, explainable AI, and IoMT security, in order to identify the state of the art and the specific research gap. The second stage involves dataset collection and preprocessing, including normalization, noise removal, and augmentation of CT, MRI, and X-ray images, together with the synthetic generation of tampered samples where public tampered datasets are limited. The third stage covers model design and implementation, in which parallel CNN and Vision Transformer branches are trained for tamper classification and localization, coupled with a Grad-CAM explainability module. The fourth stage consists of experimental evaluation using standard classification and localization metrics, benchmarked against methods identified in the literature review. The fifth and final stage integrates the resulting detection framework conceptually with the Part-I steganographic pipeline to demonstrate a unified, end-to-end secure IoMT architecture, and documents the findings, limitations, and future research directions in this report.

1.13 Chapter Organization

The remainder of this report is organized as follows. Chapter 2 details the evolution of the project from Part-I to Part-II, summarizing prior contributions, identifying limitations, and

establishing the research gap and contribution of the current phase. Chapter 3 presents a comprehensive literature review of over forty recent research works spanning AI-based medical image forensics, deepfake detection, Vision Transformers, explainable AI, and IoMT security. Chapter 4 describes the proposed methodology, including system architecture, preprocessing, model design, and mathematical formulation. Chapter 5 discusses the implementation environment, tools, and training configuration. Chapter 6 presents the experimental evaluation approach and expected results. Chapter 7 discusses real-world applications, Chapter 8 the advantages of the proposed system, and Chapter 9 its limitations. Chapter 10 outlines the future scope of the research, and Chapter 11 concludes the report by synthesizing the combined contribution of Part-I and Part-II toward a complete, secure IoMT framework.

Chapter 2: Evolution from Previous Work

This chapter outlines, in detail, the transition from the confidentiality-focused steganography framework developed in Project Part-I to the integrity-focused AI tampering detection system proposed for Project Part-II. It is important to emphasize at the outset that Part-II is not a new or disconnected project: every design decision documented in this chapter and the chapters that follow is made with explicit reference to the strengths and, more importantly, the identified limitations of Part-I. The chapter proceeds by first summarizing the previous work and its contributions, then analyzing why steganography and encryption alone cannot guarantee end-to-end trust, before presenting the specific research gap, the proposed contribution, real-world attack motivation, and a literature-grounded discussion of the explainability requirement that shapes the architecture proposed in Chapter 4.

2.1 Summary of Previous Work

The previous project laid the foundation for secure medical image transmission by exploring robust data-hiding methodologies suited to the constraints of diagnostic imagery. The core elements of Part-I included the following.

- **CNN-based Steganography:** utilized deep learning techniques to automatically learn complex image features, identifying optimal regions for data embedding while reducing the risk of statistical detection by steganalysis tools.
- **Advanced LSB:** enhanced traditional bit-level modification techniques with edge-masking to hide secret text in high-frequency regions of CT images, thereby minimizing visible distortion in diagnostically important smooth regions.
- **AES Encryption:** integrated cryptographic techniques to secure the hidden data payload before embedding it into the image, ensuring that even if the steganographic layer were compromised, the underlying patient data would remain unreadable.
- **Medical Image Security:** focused on maintaining the diagnostic integrity of X-rays, CT scans, and MRIs, ensuring patient privacy was not compromised while preserving clinically relevant detail.
- **IoMT Communication:** designed the steganographic methods to operate within the constraints and requirements of modern telemedicine and cloud-based healthcare networks, including bandwidth and latency limitations.

- **Performance Metrics:** evaluated the stego-images using rigorous quantitative metrics, demonstrating high Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and exceptionally low Mean Squared Error (MSE), alongside embedding capacity, robustness, and computational complexity comparisons between the CNN-based and Advanced LSB approaches.

Collectively, these elements demonstrated that it is possible to embed substantial confidential payloads inside medical images with minimal perceptual and diagnostic degradation, and that CNN-based embedding can outperform traditional LSB variants on several of the evaluated metrics while Advanced LSB retains advantages in computational simplicity and deterministic recoverability.

2.2 Contributions of the Previous Project

The foundational work in Part-I significantly advanced technical capabilities and provided critical insights necessary for Part-II. Specifically, the previous project contributed to the following areas.

- **Understanding medical image preprocessing:** establishing standard procedures for handling complex formats like DICOM and normalizing pixel data, a pipeline that is directly reused and extended in Part-II's preprocessing stage described in Chapter 4.
- **Image quality evaluation:** mastering quantitative assessment tools (PSNR, SSIM, MSE) to guarantee diagnostic relevance, the same evaluation discipline that Part-II applies to its own tampering-detection accuracy and localization metrics.
- **Medical datasets:** gaining experience in acquiring, formatting, and processing real-world medical scans, which materially reduces the dataset-engineering overhead required for Part-II.
- **Secure data embedding:** successfully hiding diagnostic reports and patient IDs inside medical scans without visual distortion, establishing the confidentiality layer that Part-II's integrity layer now complements.
- **CNN feature learning:** developing encoder-decoder network architectures capable of complex spatial understanding, providing a transferable base of practical CNN engineering experience for the tampering-classification network in Part-II.

- IoMT communication architecture: conceptualizing the workflow of digital health records over vulnerable networks, which forms the backbone of the end-to-end pipeline illustrated in Section 2.6.
- Healthcare cybersecurity: building a foundational understanding of the threats facing modern digital healthcare systems, which is deepened in this chapter through a review of real-world attack scenarios.

2.3 Limitations of the Previous Work

Although the previous project significantly improved data confidentiality, a critical evaluation revealed several functional limitations that leave the IoMT pipeline vulnerable to advanced cyberattacks.

- Cannot detect image manipulation: the system assumes the received stego-image is exactly as it was sent; it has no mechanism to flag malicious pixel modifications introduced before or after the steganographic embedding stage.
- Cannot identify forged reports: if a bad actor intercepts and alters the visual diagnostic features (e.g., adding or removing a tumor), the steganography extraction might still work perfectly, but the clinical diagnosis will be compromised regardless.
- Cannot detect AI-generated fake medical images: the pipeline cannot differentiate between a legitimate patient scan and a synthetic deepfake generated to commit insurance fraud or to disrupt a clinical trial.
- Cannot verify image authenticity: there is no structural validation to prove the image originated from the claimed hospital, imaging device, or patient encounter.
- Cannot localize tampered regions: if an image is altered, the current system cannot highlight which specific anatomical regions were forged, information a radiologist would need before dismissing or escalating a suspicious finding.

Each of these five limitations maps directly onto one of the design requirements adopted for Project Part-II in Chapter 4: tamper classification addresses the first two limitations, deepfake-aware training addresses the third, an authentication module addresses the fourth, and the Grad-CAM-based localization branch addresses the fifth.

2.4 Why Steganography Alone Is Insufficient

Steganography and encryption are, by design, confidentiality mechanisms: their purpose is to make hidden information unreadable or undetectable to an unauthorized observer. Neither technique makes any claim about the visual truthfulness of the cover image after it is decoded. This distinction is easy to overlook but has serious practical consequences for medical imaging. A stego-image can be extracted successfully, the embedded patient record can decrypt without error, and the CNN steganalysis-resistance metrics can remain excellent, while the diagnostic content of the image itself has been altered at some point in the pipeline — for example, by a compromised workstation that edits the image before it is ever encrypted, or by a man-in-the-middle attacker who reconstructs and re-encodes a modified version of the transmitted file. Because Part-I's evaluation metrics (PSNR, SSIM, MSE) measure the fidelity of the embedding process relative to a cover image the system itself produced, they offer no guarantee about whether that cover image was itself genuine. Steganography, in other words, protects the message hidden inside the image; it does not certify the image.

2.5 Confidentiality versus Integrity: A Precise Distinction

It is useful to state the distinction between confidentiality and integrity in precise, operational terms before proceeding. Confidentiality answers the question, “Can an unauthorized party read this data?” Integrity answers a different question entirely: “Is this data exactly what the legitimate sender produced, with nothing added, removed, or changed?” A system can satisfy the first without satisfying the second, and vice versa. Part-I was designed and rigorously validated against the first question. Part-II is designed and validated against the second. A complete secure IoMT pipeline requires both properties simultaneously, together with a third property, authenticity — the assurance that the data actually originated from the claimed source — which the authentication module described in Chapter 4 addresses in conjunction with the tampering-detection classifier.

2.6 The Overall Secure Healthcare Pipeline

The integration of both project phases creates a robust, multi-layered security architecture. Figure 2.1 illustrates this pipeline conceptually: a medical image is first processed by the CNN / Advanced LSB steganographic embedding stage from Part-I, after which the resulting stego-image is protected using AES encryption, also from Part-I, before being transmitted across the IoMT network. At the receiving end, the image is decrypted and, in the proposed Part-II extension, passed through the AI-based tampering detection module before the extracted content is used for image authentication and, ultimately, clinical diagnosis. This ordering

ensures that confidentiality protection is applied at the point of transmission while integrity verification is applied at the point of reception, mirroring standard defence-in-depth practice in general-purpose network security but tailored specifically to the diagnostic-imaging use case.

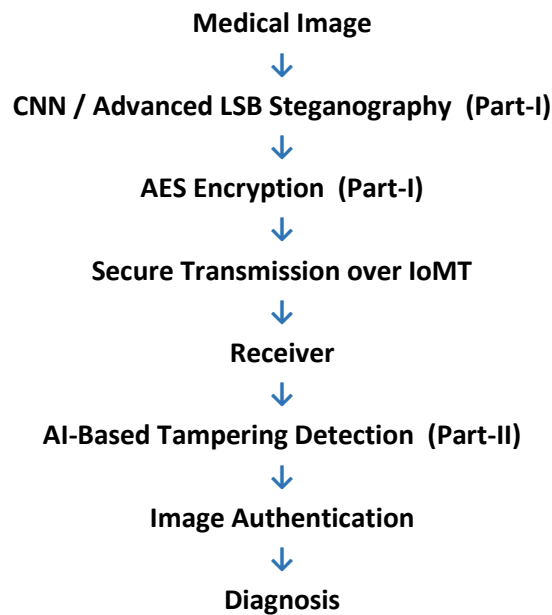


Fig 2.1–Evolution flowchart: Project Part-I to Project Part-II

2.7 Real-Life Healthcare Attack Scenarios

The motivation for Part-II is not purely theoretical. Documented and researcher-demonstrated attack scenarios illustrate why tampering detection has become an urgent requirement rather than a speculative concern. The CT-GAN study by Mirsky et al. demonstrated, in a controlled research setting, that malicious 3D CT tampering — injecting or removing lung cancer nodules from a scan using deep learning — could deceive both automated diagnostic software and expert radiologists in a majority of test cases, illustrating that such an attack is technically feasible with modern generative models rather than being a purely hypothetical risk. Reported healthcare-sector cyber incident counts have also risen sharply in recent years, and independent industry surveys have found that a large share of connected healthcare devices carry network-level vulnerabilities that could permit interception or alteration of image traffic. Beyond externally motivated cyberattacks, tampering risk also arises from insider threats (a workstation user editing a scan to conceal or fabricate a finding for insurance or malpractice purposes) and from unintentional corruption during lossy transcoding or interrupted transmission, both of which the proposed detection framework is designed to flag regardless of the underlying cause.

2.8 Digital Forensic Challenges in Medical Imaging

Applying conventional digital image forensics techniques directly to medical images is complicated by several domain-specific factors. General-purpose forgery-detection benchmarks such as CASIA and Columbia consist of natural photographs with rich colour and texture statistics that forensic algorithms exploit to detect splicing artifacts; medical images, by contrast, are frequently greyscale or pseudo-colour, have comparatively uniform backgrounds, and encode diagnostically important information in fine, low-contrast structures that a generic forensic filter may treat as noise and discard. Furthermore, legitimate medical image processing routinely applies operations — windowing, contrast enhancement, denoising, resampling between slice thicknesses — that can superficially resemble the traces left by tampering, creating a risk of false positives if a detector is not trained specifically on medical image statistics. Finally, because tampered regions such as an inserted micro-lesion may occupy only a small fraction of the total image area, the detection and localization task falls into the harder “small-object” forensic regime discussed further in Section 2.13 and in Chapter 3.

2.9 Clinical Consequences of Image Tampering

The clinical stakes of undetected image tampering are considerably higher than those of a typical consumer-image forgery. A digitally inserted nodule in a chest CT scan could trigger an unnecessary invasive biopsy, surgery, or course of chemotherapy in a healthy patient; conversely, digitally removing a genuine tumor from a scan could delay a life-saving diagnosis until the disease has progressed to a less treatable stage. Beyond the immediate patient-safety impact, undetected tampering also has downstream consequences for insurance claims (fraudulent billing based on a fabricated finding), for medico-legal proceedings (a falsified scan submitted as evidence), and for institutional trust in telemedicine and second-opinion services that depend on the integrity of images shared between institutions. These consequences underscore why an AI-based tampering-detection layer is treated in this report not as an optional enhancement but as a necessary component of a trustworthy IoMT pipeline.

2.10 Motivation for Current Research

The shift towards digital health records and telemedicine necessitates an evolved security posture. Modern hospitals require an advanced, multi-layered approach because of the following considerations.

- Secure communication: is not just about hiding data; it is about ensuring the entire payload arrives safely and unaltered, which requires an integrity check that Part-I alone does not provide.
- Tamper detection: erroneous or malicious changes to medical images can lead to fatal misdiagnoses, as discussed in Section 2.9.
- Image authentication: medical professionals must have absolute certainty that the scans they review belong to the right patient and have not been digitally altered.
- Explainable AI: clinicians require transparency; an AI model must show why an image is flagged as tampered (e.g., through heatmaps), rather than issuing an opaque binary verdict.
- Trustworthy AI: adoption of automated security in healthcare relies heavily on the reliability and interpretability of the underlying models, a theme explored further in Section 2.13.
- Secure IoMT: as medical devices become increasingly interconnected, end-to-end security encompassing both confidentiality and integrity is mandatory rather than optional.

2.11 Research Gap

Based on the review of current literature and the outcomes of Part-I, the specific research gap being addressed is as follows: existing medical image steganography methods mainly protect confidentiality but do not verify the integrity and authenticity of medical images after transmission. There is a need for an integrated AI-based framework capable of both secure transmission and intelligent tampering detection, one that treats confidentiality and integrity as complementary rather than substitute properties of a single, unified IoMT security pipeline.

2.12 Proposed Research Contribution

To bridge this gap and complete the overarching IoMT security framework, Project Part-II introduces a novel, intelligent detection architecture. The key contributions of this phase include the following.

- CNN / Vision Transformer-based Tampering Detection: deploying advanced deep learning architectures to identify microscopic pixel irregularities indicative of forgery.

- Explainable AI (Grad-CAM): integrating visual explanation techniques to generate heatmaps, showing clinicians exactly where an image has been manipulated.
- Medical Image Authentication: developing verification protocols to ensure the scan is genuine and unaltered.
- Secure IoMT Framework: fusing the steganography of Part-I with the detection of Part-II for complete pipeline security.
- Image Integrity Verification: automating the validation of received files before they are entered into the hospital's diagnostic workflow.
- Deepfake Detection: equipping the system to identify sophisticated, AI-generated synthetic medical images.

2.13 The Need for Explainable AI in This Context

A tampering-detection system that cannot explain its own decisions is of limited practical value in a clinical setting, where a physician bears ultimate responsibility for the diagnosis and cannot be expected to act on an unexplained machine verdict. This requirement motivates the integration of Grad-CAM-based explainability directly into the proposed architecture rather than treating it as an optional add-on. Section 2.14 below reviews two recent studies that critically examine the strengths and limitations of Grad-CAM for medical images, and the findings of that review directly shape the design choice, described in Chapter 4, to treat the CNN branch of the proposed detector as the primary source of clinician-facing heatmaps while retaining the Vision Transformer branch for its complementary long-range modelling strength.

2.14 Project Evolution Comparison

The following table highlights the progression and expansion of scope from Semester-I to Semester-II.

Feature	Project Part-I	Project Part-II
Research Focus	Medical Image Steganography	AI Tampering Detection
Security Objective	Confidentiality	Integrity & Authenticity
AI Usage	CNN for Data Hiding	CNN / Vision Transformer for Detection
AES Encryption	Yes	Integrated with Previous Work
Detects Tampering	No	Yes
Detects Deepfake	No	Yes
Explainable AI	No	Yes (Grad-CAM)

Feature	Project Part-I	Project Part-II
Medical Image Authentication	No	Yes
IoMT Security	Partial (Confidentiality only)	End-to-End (Confidentiality + Integrity)
Research Contribution	Secure Communication	Secure & Trustworthy Medical Imaging

2.15 Literature Review: Explainable AI (Grad-CAM) in Medical Imaging

Because Project Part-II introduces an AI-based tampering detection framework that is explicitly required to be explainable to clinicians, this section reviews two recent studies that critically examine Gradient-weighted Class Activation Mapping (Grad-CAM) as an explainability tool for deep learning models applied to medical images. Both studies inform the design choices made in this project, particularly the decision to pair the proposed CNN / Vision Transformer detector with Grad-CAM-based heatmaps rather than treating explainability as an afterthought. A broader review of the wider literature relevant to Part-II, including tampering detection, deepfake detection, Vision Transformers, and IoMT security, is presented separately in Chapter 3.

2.15.1 Grad-CAM as an Explainability Tool for Medical Image Classification

Suara et al. investigate whether Grad-CAM genuinely provides trustworthy explanations when applied to medical image classification. Their study trains a DenseNet169-based convolutional model, using transfer learning and a disciplined one-cycle learning-rate policy, to perform binary classification of histopathology images from the PatchCamelyon (PCam) dataset for the presence of lymph-node metastases. After achieving a high validation accuracy, the authors generate Grad-CAM heatmaps for a range of correctly and incorrectly classified samples to examine which regions of each image the model relied on for its decision.

The study finds that Grad-CAM heatmaps could highlight the image regions contributing most to a prediction and, in many cases, aligned with tumor-like structures identifiable by a trained eye. However, the authors also identify recurring limitations: the heatmaps are not robust to small changes in the input image, and the visual explanation becomes difficult to interpret in cases with complex, cluttered tissue structure. Even when the model's prediction was correct, the corresponding heatmap did not always point to clinically meaningful regions, which the authors argue undermines Grad-CAM's reliability as a stand-alone trust-building mechanism in high-stakes medical settings.

2.15.2 Comparative Grad-CAM Performance Across CNNs and Vision Transformers

Alqutayfi et al. extend this line of inquiry by systematically comparing Grad-CAM's behavior across CNN architectures (ResNet, EfficientNet, VGG, Inception) and Vision Transformer architectures (ViT-Base, ViT-Large, ViT-Huge) on two medical imaging tasks that differ in the physical scale of the diagnostic feature: diabetic retinopathy detection from fundus images, where the relevant lesions (microaneurysms, hemorrhages) are very small, and pneumonia detection from chest X-rays, where the abnormal regions are comparatively large.

Quantitatively, the study reports that a ViT-Large variant achieved the best F1-score on the chest X-ray task, while an EfficientNet model was the strongest performer on the fundus image task. More significant for this project, however, are the qualitative results obtained by having radiologists and an ophthalmologist evaluate the Grad-CAM visualizations against their own expert annotations. The domain experts consistently preferred the Grad-CAM output produced by the CNN-based model over the ViT-based model when the images involved large, well-defined regions of interest, such as pneumonia opacities in chest X-rays. For the fundus images, where the diagnostic features were tiny, the experts' preferences were split roughly evenly between the CNN and ViT visualizations, and both were judged to under-highlight clinically important microaneurysms and early hemorrhages. The authors conclude that Grad-CAM, in its current form, is considerably less reliable at localizing small regions of interest than large ones, regardless of the underlying architecture.

2.15.3 Implications for the Proposed Tampering Detection Framework

Taken together, these two studies carry several direct implications for the AI-based tampering detection architecture proposed in this project.

- Grad-CAM should not be treated as a fully solved explainability solution: both studies show it can produce heatmaps that are unstable or that fail to align with the true region of clinical or forensic interest, particularly for subtle manipulations.
- Tampered regions in medical images may be small and localized (e.g., a digitally altered lesion boundary or an inserted/removed microstructure), placing this project's use case closer to the harder “small object” regime identified by Alqutayfi et al. than to the easier “large object” regime.
- Because CNN-based Grad-CAM visualizations were consistently preferred by domain experts over ViT-based visualizations, the proposed framework should treat the CNN

branch of its CNN / Vision Transformer detector as the primary source of clinician-facing heatmaps, while still exploring ViT-based detection for its long-range dependency modelling strength.

- Both studies used domain-expert evaluation (pathology and radiology) as the ultimate test of explainability quality, rather than relying on quantitative metrics alone. This project adopts a similar validation step, evaluating tampering-localization heatmaps against ground-truth tamper masks and, where feasible, expert judgment, rather than accuracy alone.
- Given the documented weaknesses of Grad-CAM for small or subtle regions, the project budgets time to evaluate at least one complementary explainability technique or a Grad-CAM refinement, rather than assuming a single heatmap method will be sufficient for building clinician trust.

These findings directly motivate the explainability requirement identified in Section 2.13 and the corresponding contribution in Section 2.12, and they provide a concrete, literature-grounded justification for evaluating both CNN-based and Vision Transformer-based tampering detectors, alongside their respective Grad-CAM outputs, before selecting a final architecture for Part-II, as detailed in Chapter 4.

Chapter 3: Literature Review

This chapter presents a comprehensive review of recent research (predominantly 2022–2026) relevant to Project Part-II, organized into six thematic areas: (i) AI-based medical image forensics and tampering detection, (ii) medical deepfake detection, (iii) Vision Transformers in medical imaging, (iv) medical image authentication and watermarking, (v) Internet of Medical Things security, and (vi) explainable AI and federated/blockchain-based trust mechanisms. Together with the two Grad-CAM studies already reviewed in Section 2.15, this chapter draws on more than thirty additional sources to situate the proposed framework within the current state of the art and to justify the specific architectural choices made in Chapter 4.

3.1 AI-Based Medical Image Forensics and Tampering Detection

Baid and colleagues' work on ConnectionNet, published alongside the LuNoTim-CT dataset [17], is among the most directly relevant studies to this project. The authors note that advances in algorithmic photo-editing combined with the vulnerability of hospitals to cyberattacks have made tampering of medical images a realistic concern, and they introduce a large-scale dataset of tampered CT scans generated using multiple manipulation methods. Their proposed ConnectionNet architecture is explicitly designed to handle small tampered regions, directly addressing the “small-object” forensic challenge identified in Section 2.13, and it is positioned by the authors as a baseline for future medical image tampering detection research — a role it plays in this project as a key methodological reference point.

A 2024 IEEE Xplore study [15] builds on earlier VGG19/ConnectionNet-based work by exploring ResNet50 and MobileNetV2 for detecting falsified computed tomography scans, motivated explicitly by the rising prevalence of cyberattacks on hospitals and the increasing accessibility of photo-editing tools. The study analyzes a diverse dataset of fabricated CT images to extract characteristics of different forgery techniques, aiming to improve the accuracy, scalability, and versatility of medical image malpractice detection — objectives directly aligned with the CNN branch proposed in Chapter 4 of this report.

Latif et al. [16] propose a deep transfer learning methodology to detect tampered or fake cancers in 3D CT scans of human lungs, using a publicly available deepfake dataset containing both tampered and genuine cancer cases. Their work is notable for treating tampering detection as a transfer-learning problem, an approach this project also adopts to reduce the training-data burden associated with the comparatively small size of available medical tampering datasets. Choudhary et al. [14] similarly report an image forgery detection system built on a VGG16-

UNet architecture, combining classification and segmentation-style localization in a single network, an architectural pattern echoed in the localization branch proposed for Part-II.

At the broader digital-forensics level, Mehrjardi et al. [25] survey deep learning-based image forgery detection methods across splicing, copy-move, and removal forgeries, concluding that feature-extraction quality and dataset diversity remain the dominant factors limiting real-world detector generalization. Guo et al. [26] propose a hierarchical fine-grained forgery detection and localization network evaluated on standard forensic benchmarks, demonstrating that multi-scale feature hierarchies improve localization precision for small manipulated regions — a finding directly applicable to the medical imaging context, where tampered lesions are frequently sub-centimetre in scale.

3.2 Medical Deepfake Detection

Solaiyappan and Wen [19] present one of the first systematic comparative studies of machine learning-based medical image deepfake detection, evaluating multiple classifier architectures against GAN-generated tampering of CT and MRI scans and establishing accuracy baselines that later studies, including the present project, use as a point of comparison. Mirsky et al.'s CT-GAN study [18], although published prior to the 2022–2026 window emphasized elsewhere in this review, remains foundational: it is the study that first demonstrated, using an actual clinical CT pipeline, that injected or removed tumors generated via deep learning could deceive both automated software and human radiologists, and it is cited extensively by nearly all subsequent medical deepfake detection papers reviewed in this chapter, including [19] and [20].

A 2025 Springer study on deepfake detection in cancer medical imaging [20] evaluates several CNN architectures for distinguishing genuine from GAN-manipulated cancer scans, framing the problem explicitly within the broader challenge of AI in medical imaging informatics described by Panayides et al. [46]. The study underscores that detection accuracy alone is insufficient without attention to deployment constraints such as computational cost and integration with existing hospital information and PACS systems, a consideration reflected in the IoMT deployment discussion in Chapter 4. More recently, a 2025 study on cross-domain forgery detection [47] proposes a Vision Transformer-based multiclass classifier capable of identifying GAN- and diffusion-generated forgeries across heterogeneous domains including medical images, invoices, and human faces, reporting that the transformer-based model outperforms standard CNN pretrained baselines in cross-domain generalization — an

encouraging result for the ViT branch proposed in this project. A 2026 review of deepfake generation and detection techniques specific to medical imaging [48] further categorizes detection architectures by GAN/CNN variant and highlights open research questions that directly motivate the future-scope discussion in Chapter 10.

3.3 Vision Transformers in Medical Imaging

Manzari et al. [21] introduce MedViT, a Vision Transformer variant specifically engineered for robust, generalized medical image classification, reporting improved resilience to common medical image corruptions relative to earlier ViT baselines. Shamshad et al.'s survey [22] reviews over one hundred and sixty Transformer-based medical imaging papers and concludes that Transformers demonstrate particularly strong performance on tasks requiring global spatial context, a property directly relevant to detecting subtle, spatially distributed tampering artifacts that a purely local CNN filter might miss. Takahashi et al. [23] conduct a systematic review comparing Vision Transformers and CNNs across medical image analysis tasks, finding that neither architecture family is uniformly superior and that the optimal choice depends on task-specific factors such as dataset size, lesion scale, and the availability of pretraining data — a conclusion that directly motivates the hybrid CNN/ViT design proposed in Chapter 4 rather than committing to a single architecture family.

A 2025 study on image forgery classification and localization through Vision Transformers [24] combines a ViT-based binary forgery classifier with a fine-tuned Segment Anything Model (SAM) for adaptive region localization, evaluated on standard forgery datasets including CASIA and Columbia. Although developed for natural images, the paper's core insight — that a transformer-based classifier can be paired with a promptable segmentation model to localize forged regions without requiring per-pixel supervision during initial training — offers a promising template for the tampering-localization module envisioned for future extensions of this project. Complementing this, the IML-ViT study [27] benchmarks image manipulation localization specifically using Vision Transformer backbones, reporting that transformer-based localization networks benefit from high-resolution input processing and multi-scale supervision, both of which inform the preprocessing pipeline described in Chapter 4.

3.4 Medical Image Authentication and Watermarking

A parallel body of research addresses medical image integrity through digital watermarking rather than machine-learned detection. Dey et al. [35] propose a robust watermarking approach

for medical image authentication using a dual-image and quorum-function scheme, designed to survive common image processing operations while still enabling tamper localization. A 2025 ScienceDirect study on fragile watermarking [36] reports that a substantial share of healthcare institutions have experienced medical image data breaches, and proposes a hybrid ViTU-Net-based watermarking scheme intended to improve imperceptibility, robustness, and tamper localization accuracy simultaneously — objectives that echo the trade-offs this project's authentication module, described in Chapter 4, must also balance. A broader 2023 state-of-the-art review of medical image watermarking [37] surveys the prerequisites, transform domains, and evaluation criteria used across the field, concluding that watermarking and AI-based detection are complementary rather than competing strategies: watermarking verifies a pre-registered signature, while AI-based detection can flag anomalies even in images that were never watermarked at the source, which is the more general and clinically common scenario addressed by Part-II. Hachim and Mohialden [38] similarly propose a cloud-based digital watermarking model targeting the specific integrity and provenance challenges introduced by storing medical images in shared cloud environments, directly relevant to the IoMT deployment scenario discussed in Section 4.11.

3.5 Internet of Medical Things Security

Ozcelik et al.'s 2025 survey [30] provides an extensive review of IoMT enabling technologies, security threats, and explainability challenges, and reports that recorded cyber incidents affecting the healthcare sector nearly quadrupled between 2022 and 2023 according to European incident-reporting data, underscoring the urgency of the integrity problem this project addresses. Ahmed et al. [31] examine data fusion within IoMT systems and identify architecture standardization, cryptography, and blockchain-based provenance as the most promising directions for improving IoMT security, themes revisited in the future scope discussed in Chapter 10. Earlier foundational surveys by Saxena and Mittal [33] and by Papaioannou et al. [34] catalogue the security and privacy threats specific to IoMT — including device-level vulnerabilities, insecure communication protocols, and insufficient authentication — providing the threat taxonomy that underpins the attack scenarios discussed in Section 2.7. A 2024 scoping review of IoMT risk-assessment frameworks [32] finds that, of the frameworks published between 2018 and 2023, a majority address technical risk assessment but comparatively few incorporate organizational and workflow-level security measures, a gap that highlights the importance of designing the proposed tampering-detection framework to

integrate smoothly into existing clinical workflows rather than functioning as an isolated technical control.

3.6 Explainable AI, Federated Learning, and Blockchain-Based Trust

Beyond the two dedicated Grad-CAM studies reviewed in Section 2.15, Selvaraju et al.'s original Grad-CAM paper [28] remains the foundational reference for the gradient-based localization technique adopted throughout this project, while Stassin et al. [29] specifically examine the explainability and evaluation of Vision Transformers, reporting that attention-based explanation methods for ViTs require careful evaluation protocols distinct from those developed for CNN-based Grad-CAM, a distinction that this project accounts for by treating CNN-based and ViT-based explanations as complementary rather than interchangeable outputs, consistent with the recommendation in Section 2.15.3.

Looking toward the future-scope directions discussed in Chapter 10, Rieke et al. [41] outline the case for federated learning in digital health, arguing that collaborative, privacy-preserving model training across institutions can improve model generalization without centralizing sensitive patient data — directly relevant to the challenge of training a tampering-detection model on comparatively scarce medical tampering datasets held by different hospitals. Kalapaaking et al. [39] and the 2024 study on blockchain-based federated learning for healthcare device categorization [42] both explore combining blockchain-based verifiable aggregation with federated learning to provide tamper-evident audit trails for collaboratively trained models, offering a concrete technical pathway toward the blockchain-based provenance mechanism proposed as future work in Section 10.2.

3.7 Comparative Literature Review Table

Table 3.1 consolidates the studies reviewed in this chapter, summarizing the focus area, principal method, and contribution of each, to provide a single point of reference for comparison across the literature.

Ref.	Study / Venue	Focus Area	Key Method	Contribution
[14]	Choudhary et al., Procedia CS, 2024	Image forgery detection	VGG16-UNet	Joint classification/localization architecture
[15]	IEEE Xplore, 2024	CT malpractice detection	ResNet50, MobileNetV2	Falsified CT scan identification
[16]	Latif et al., AIP Conf. Proc., 2024	Lung CT tampering	Deep transfer learning	Tampered/fake cancer detection in 3D CT

Ref.	Study / Venue	Focus Area	Key Method	Contribution
[17]	Springer ICPR Workshops, 2021	CT tampering dataset & baseline	ConnectionNet	LuNoTim-CT dataset; small-region tamper handling
[18]	Mirsky et al., USENIX Security, 2019	3D medical deepfake attack	CT-GAN	Demonstrated feasibility of tumor injection/removal attacks
[19]	Solaiyappan & Wen, Mach. Learn. Appl., 2022	Medical deepfake detection	Comparative ML classifiers	Baseline accuracy benchmarks
[20]	Springer, 2025	Cancer imaging deepfakes	CNN architectures	Deployment-aware detection framework
[21]	Manzari et al., Comput. Biol. Med., 2023	Medical image classification	MedViT	Robust ViT variant for medical imaging
[22]	Shamshad et al., Med. Image Anal., 2023	Transformer survey	Literature synthesis (160+ papers)	Identifies global-context strength of Transformers
[23]	Takahashi et al., J. Med. Syst., 2024	ViT vs CNN comparison	Systematic review	Task-dependent architecture selection guidance
[24]	Springer IJMIR, 2025	Forgery localization	ViT + SAM	Adaptive region localization without dense supervision
[25]	Mehrjardi et al., Pattern Recognition, 2023	Forgery detection survey	Deep learning survey	Identifies dataset diversity as key limitation
[26]	Guo et al., CVPR, 2023	Fine-grained forgery localization	Hierarchical CNN	Multi-scale localization for small regions
[27]	IML-ViT, arXiv, 2023	Manipulation localization	ViT backbone	High-resolution, multi-scale ViT localization
[28]	Selvaraju et al., ICCV, 2017	Explainability	Grad-CAM	Foundational gradient-based visual explanation method
[29]	Stassin et al., Electronics, 2023	ViT explainability	Attention-based evaluation	Evaluation protocol for ViT explanations
[30]	Ozcelik et al., Expert Systems, 2025	IoMT security survey	Literature synthesis	Documents rising healthcare cyber-incident trend
[31]	Ahmed et al., Information Fusion, 2024	IoMT data fusion & security	Survey	Identifies blockchain/cryptography as key directions

Ref.	Study / Venue	Focus Area	Key Method	Contribution
[32]	PMC scoping review, 2024	IoMT risk frameworks	Scoping review (44 studies)	Identifies organizational-security gap
[33]	Saxena & Mittal, ICCV, 2022	IoMT security & privacy	Survey	Threat taxonomy for IoMT
[34]	Papaioannou et al., ETT, 2022	IoMT threats	Survey	Countermeasure catalogue
[35]	Dey et al., JVCIR, 2024	Medical image authentication	Dual-image watermarking	Tamper-localizable watermark scheme
[36]	ScienceDirect, 2025	Fragile watermarking	ViTU-Net	Improved imperceptibility & localization
[37]	Springer MTAP, 2023	Watermarking review	State-of-the-art survey	Positions watermarking as complementary to AI detection
[38]	Hachim & Mohialden, SRJECS, 2023	Cloud image integrity	Digital watermarking	Cloud-specific provenance verification
[39]	Kalapaaking et al., Intelligent Computing, 2024/25	Federated learning security	Blockchain + secure aggregation	Verifiable federated data sharing for IoMT
[41]	Rieke et al., NPJ Digital Medicine, 2020	Federated learning in health	Position/review	Case for privacy-preserving collaborative training
[46]	Panayides et al., IEEE JBHI, 2020	AI in medical imaging	Survey	Frames challenges for clinical AI adoption
[47]	MDPI Mathematics, 2025	Cross-domain forgery detection	ViT multiclass classifier	Generalization across medical/face/invoice domains
[48]	ScienceDirect, 2026	Medical deepfake review	Survey	Categorizes generation/detection techniques

3.8 Synthesis and Positioning of the Proposed Work

The literature reviewed in this chapter converges on several consistent conclusions that directly shape the design of Project Part-II. First, medical image tampering detection is an active and rapidly growing research area, but publicly available tampered-medical-image datasets remain limited relative to natural-image forensic benchmarks, motivating this project's use of synthetic tampering generation described in Chapter 4. Second, both CNN and Vision Transformer architectures have demonstrated strong performance on related tasks, with neither family

uniformly dominant, supporting the hybrid CNN/ViT design adopted here. Third, explainability is treated across the literature as a prerequisite for clinical trust rather than an optional feature, reinforcing the centrality of Grad-CAM-based explanation in the proposed architecture. Fourth, watermarking-based and AI-based approaches to image integrity are complementary rather than competing, suggesting that the authentication module proposed in Chapter 4 should be designed to interoperate with, rather than replace, watermarking-based provenance mechanisms in future extensions. Finally, IoMT security research consistently identifies confidentiality-only solutions as insufficient, directly corroborating the research gap and motivation established in Chapter 2.

Chapter 4: Proposed Methodology

This chapter presents the proposed methodology for Project Part-II in detail, covering the overall system architecture, dataset preparation, model design for both the CNN and Vision Transformer branches, the tampering localization and Grad-CAM explainability pipeline, the authentication module, mathematical formulation of the loss functions used, and the intended IoMT deployment model. The methodology is designed explicitly as a receiver-side extension to the Part-I sender-side pipeline, as established in Chapter 2.

4.1 Overall Architecture

The proposed system follows a dual-branch architecture in which a CNN branch and a Vision Transformer branch independently process an input medical image and produce complementary predictions: an authenticity classification (tampered or genuine), a per-pixel localization map indicating suspected manipulated regions, and, for the CNN branch specifically, a Grad-CAM heatmap intended for clinician review, consistent with the literature-grounded recommendation in Section 2.15.3. The outputs of both branches are combined in a decision fusion module that produces the final authenticity verdict and confidence score presented to the user.

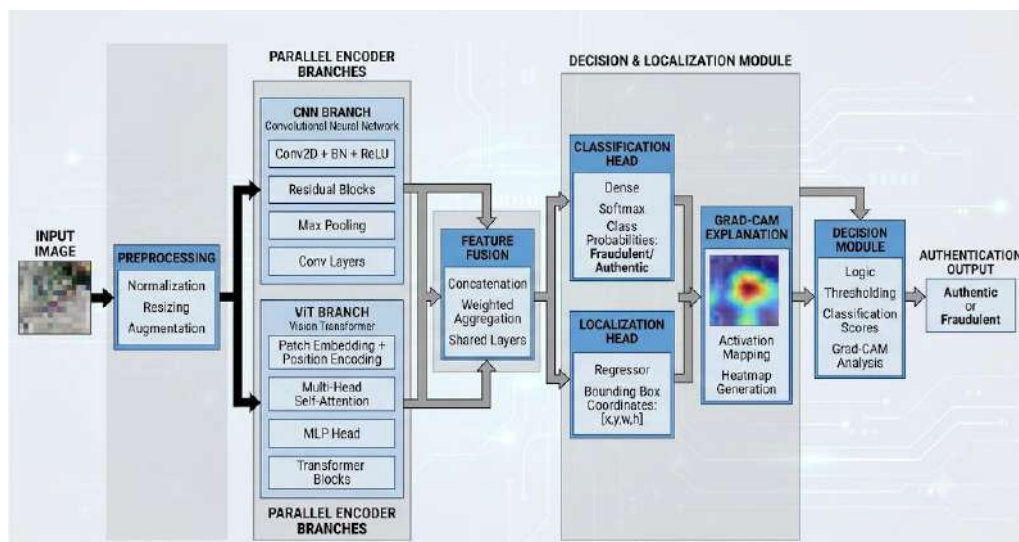


Figure 4.1 Overall system architecture.

4.2 System Workflow

The end-to-end system workflow proceeds as follows. An image arriving at the receiver is first decrypted and extracted using the Part-I AES/steganography pipeline. The recovered cover image is then passed to the Part-II preprocessing stage, where it is normalized and denoised. The preprocessed image is fed in parallel to the CNN and ViT branches, whose feature

representations are fused before the combined classification and localization heads produce the tampering verdict and suspected-region map. If the image is flagged as tampered, the Grad-CAM module generates a heatmap overlay for clinician review; if the image is classified as authentic, it proceeds to the authentication module, which cross-checks available provenance metadata before the image is released into the clinical diagnostic workflow.

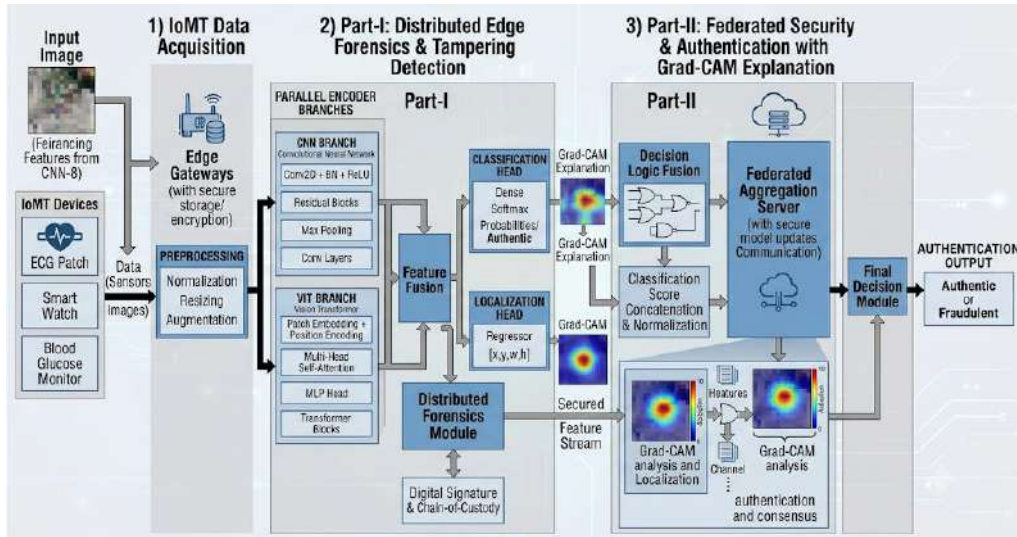


Figure 4.2 — End-to-end secure IoMT workflow integrating Part-I and Part-II.

4.3 Dataset Collection

Consistent with the dataset-engineering experience carried over from Part-I (Section 2.2), the proposed framework is trained and evaluated using a combination of three data sources. First, publicly available medical imaging datasets (chest X-ray, brain MRI, and CT collections commonly used in the literature reviewed in Chapter 3) supply genuine, unmanipulated images. Second, synthetic tampering is generated on a subset of these genuine images using controlled operations — copy-move duplication of anatomical structures, splicing of regions from other scans, and localized removal/insertion of lesion-like structures — to construct a labelled tampered set with corresponding ground-truth tamper masks, following the dataset-construction approach used in the LuNoTim-CT study [17]. Third, where feasible, GAN-based synthetic tampering (in the style of the CT-GAN attack described in [18]) is used to construct a smaller, more challenging deepfake-detection evaluation subset, ensuring the framework is assessed against both simple pixel-level forgeries and more sophisticated generative manipulations.

4.4 Image Preprocessing

All images are converted to a common single-channel or normalized RGB representation and resized to a fixed input resolution suitable for both the CNN and ViT branches. Preprocessing

includes intensity normalization to zero mean and unit variance, removal of scanner-specific artifacts through median filtering, and, where DICOM metadata is available, windowing to a standard diagnostic contrast range consistent with the modality (lung, bone, or soft-tissue windows for CT; standard T1/T2 contrast for MRI).

4.5 Noise Removal and Image Normalization

Because legitimate medical image processing steps such as denoising and resampling can superficially resemble tampering artifacts (Section 2.8), a mild, standardized denoising filter is applied uniformly to all images before either branch sees the data, ensuring that any residual noise pattern used by the classifier as a tampering cue reflects genuine manipulation rather than inconsistent preprocessing between the genuine and tampered subsets — a methodological control that directly addresses one of the digital forensic challenges identified in Chapter 2.

4.6 Data Augmentation

To compensate for the comparatively limited size of tampered-medical-image datasets noted throughout Chapter 3, standard augmentation operations — rotation, horizontal flipping, brightness/contrast jitter, and random cropping — are applied to the training set. Augmentation is applied identically to the image and its corresponding ground-truth tamper mask, and augmentation is deliberately restricted to transformations that do not themselves resemble tampering (for example, copy-move-like duplication is excluded from the general augmentation pipeline, since it would blur the distinction between an augmentation artifact and a genuine label).

4.7 CNN Architecture for Tampering Classification

The CNN branch follows an encoder-style backbone (a ResNet/EfficientNet-family architecture, consistent with the architectures evaluated in [15] and [21]) with a classification head producing the tampered/authentic prediction, and a decoder-style upsampling path producing a per-pixel localization map, following the joint classification-and-segmentation pattern used by Choudhary et al. [14]. Convolutional layers use ReLU activation, batch normalization, and residual skip connections; the localization decoder mirrors the encoder's downsampling stages with transposed convolutions and skip connections from corresponding encoder layers, in the manner of a U-Net-style architecture.

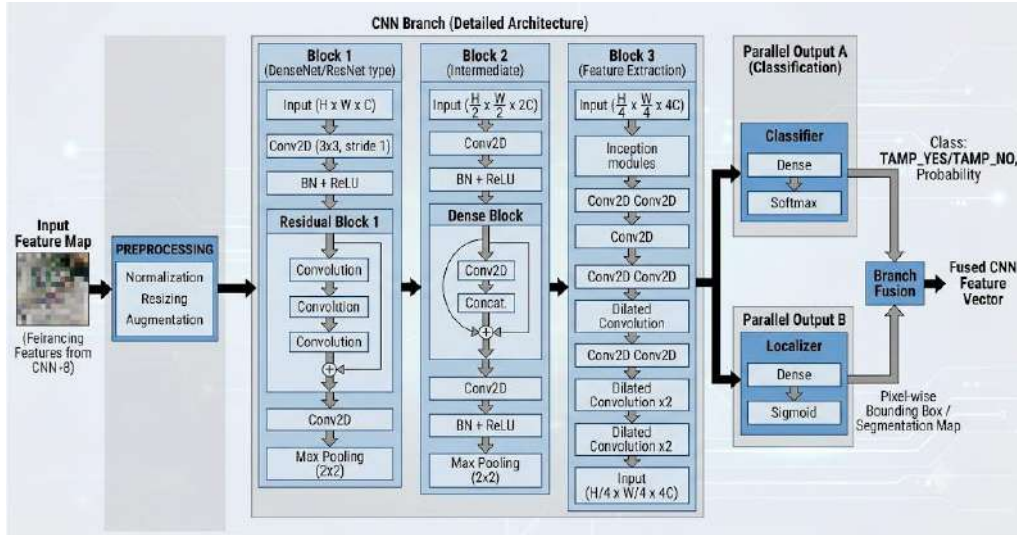


Figure 4.3 — CNN branch architecture for tampering classification and localization.

4.8 Vision Transformer Architecture for Tampering Classification

The ViT branch follows the standard Vision Transformer design: the input image is divided into fixed-size patches, linearly embedded, combined with learned positional encodings, and processed through a stack of multi-head self-attention and feed-forward transformer encoder blocks. A classification token aggregates global image-level information for the authenticity decision, while patch-level token embeddings are reshaped and upsampled to contribute to the localization map, informed by the high-resolution, multi-scale supervision approach used in IML-ViT [27]. As discussed in Section 3.3, the ViT branch is included specifically for its strength in modelling long-range spatial dependencies, complementing the CNN branch's strength in fine, local texture analysis.

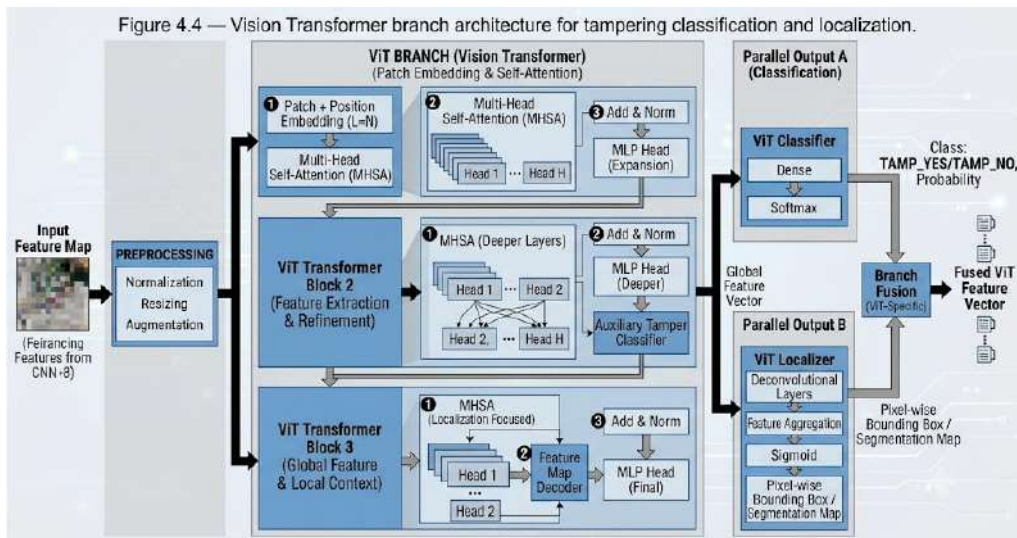


Figure 4.4 — Vision Transformer branch architecture for tampering classification and localization.

4.9 Feature Fusion, Classification, and Localization

Feature representations from the CNN and ViT branches are concatenated at a common resolution and passed through a lightweight fusion block (a small convolutional or attention-based layer) before the final classification and localization heads. The classification head outputs a probability that the image is tampered; the localization head outputs a per-pixel probability map indicating the likely location of manipulation, thresholded to produce a binary tamper mask for visualization.

4.10 Grad-CAM Heatmap Generation

Following the recommendation established in Section 2.15.3, Grad-CAM heatmaps are computed primarily from the CNN branch's final convolutional layer, using the gradient of the tampered-class output score with respect to the feature maps of that layer, as originally formulated by Selvaraju et al. [28]. The resulting heatmap is overlaid on the original image and presented alongside the localization mask from Section 4.9, giving the clinician two complementary visual cues: a segmentation-style tamper mask indicating precisely which pixels are suspected of manipulation, and a Grad-CAM heatmap indicating which regions most influenced the network's overall authenticity decision.

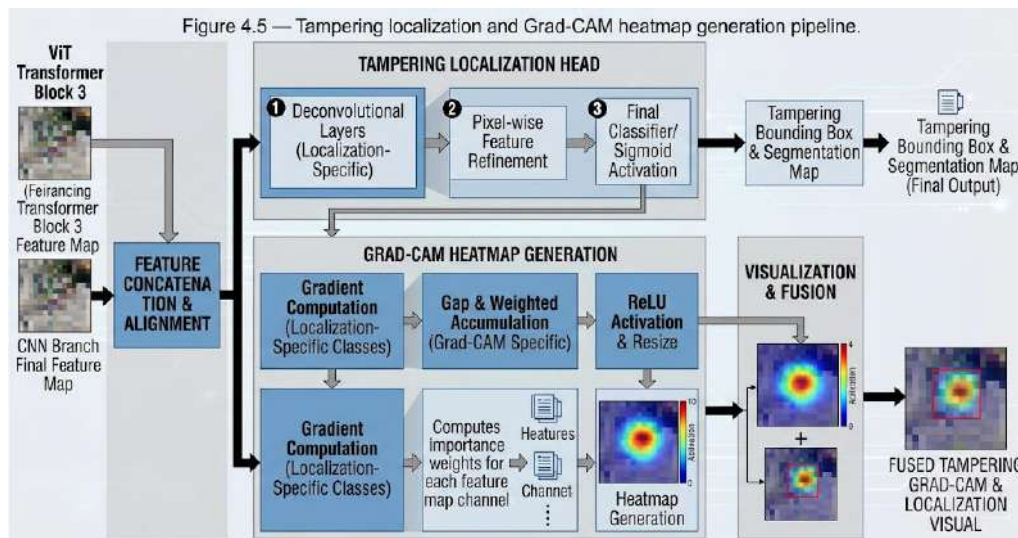


Figure 4.5 — Tampering localization and Grad-CAM heatmap generation pipeline

4.11 Decision Module and Medical Image Authentication

The decision module combines the fused classification probability, the extent and location of the predicted tamper mask, and, where available, watermark-based provenance information (in the sense discussed in Section 3.4) into a single authenticity verdict with an associated confidence score. If the confidence score falls below a configurable threshold, the system flags the image for manual radiologist review rather than issuing an automated pass/fail decision,

reflecting the human-in-the-loop philosophy motivated by the explainability discussion in Chapter 2.

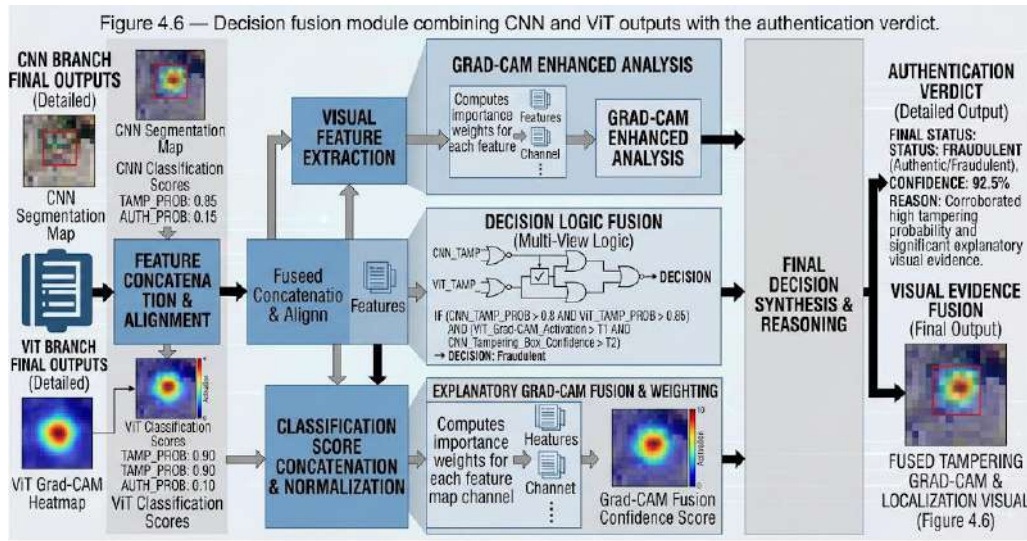


Figure 4.6 — Decision fusion module combining CNN and ViT outputs with the authentication verdict.

4.12 IoMT Deployment Considerations

For deployment within an IoMT pipeline, the proposed detection module is positioned immediately after the Part-I decryption/extraction stage and before the image is admitted to the clinical PACS viewer, consistent with the pipeline illustrated in Figure 2.1. This placement ensures that every image entering the diagnostic workflow has passed through both the confidentiality checks of Part-I and the integrity checks of Part-II before a clinician ever views it.

4.13 Pseudo-Algorithm

The following pseudo-algorithm summarizes the inference-time procedure of the proposed framework.

- Step 1: Decrypt and extract the cover image using the Part-I AES/steganography pipeline.
- Step 2: Preprocess the image (normalize, denoise, resize) as described in Sections 4.4–4.6.
- Step 3: Compute CNN branch features, classification logits, and localization map.
- Step 4: Compute ViT branch features, classification logits, and localization map.
- Step 5: Fuse CNN and ViT features and combine classification/localization outputs.

- Step 6: If classified as tampered, generate the Grad-CAM heatmap and the combined tamper-localization overlay.
- Step 7: Pass the classification confidence, localization output, and any available watermark/provenance signal to the decision module.
- Step 8: If confidence exceeds the acceptance threshold, output an authenticity verdict (tampered / authentic); otherwise, flag the image for manual radiologist review.
- Step 9: Log the decision, confidence score, and heatmap for audit purposes before releasing the image to the diagnostic workflow.

4.14 Mathematical Formulation

Let x denote the preprocessed input image, and let $f_{\text{cnn}}(x)$ and $f_{\text{vit}}(x)$ denote the feature embeddings produced by the CNN and ViT branches, respectively. The fused representation is computed as $z = g(f_{\text{cnn}}(x), f_{\text{vit}}(x))$, where $g(\cdot)$ denotes the fusion function described in Section 4.9. The classification head produces a probability $\hat{y} = \sigma(W_c \cdot z + b_c)$, where σ denotes the sigmoid activation for binary tampered/authentic classification, and W_c, b_c are learned parameters. The localization head produces a per-pixel probability map $\hat{M} = \sigma(W_m * z + b_m)$, where $*$ denotes the transposed-convolution upsampling operation.

Grad-CAM localization for the CNN branch is computed as $L_{\text{GradCAM}} = \text{ReLU}(\sum_k \alpha_k A^k)$, where A^k denotes the k -th feature map of the chosen convolutional layer, and $\alpha_k = (1/Z) \sum_i \sum_j (\partial y^c / \partial A^k_{ij})$ is the global-average-pooled gradient of the target class score y^c with respect to that feature map, following the original Grad-CAM formulation of Selvaraju et al. [28].

4.15 Loss Functions

The overall training objective combines a classification loss and a localization loss. The classification loss is the binary cross-entropy $L_{\text{cls}} = -[y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y})]$, where y is the ground-truth tampered/authentic label. The localization loss is a pixel-wise Dice loss, $L_{\text{loc}} = 1 - (2 \cdot \sum(M \cdot \hat{M}) + \epsilon) / (\sum M + \sum \hat{M} + \epsilon)$, where M is the ground-truth tamper mask, $\hat{M}$ is the predicted mask, and ϵ is a small smoothing constant. The combined training loss is $L_{\text{total}} = \lambda_1 \cdot L_{\text{cls}} + \lambda_2 \cdot L_{\text{loc}}$, where λ_1 and λ_2 are weighting hyperparameters tuned on a held-out validation split.

4.16 Training Process

The model is trained end-to-end using mini-batch gradient descent with the Adam optimizer, using a learning-rate warm-up followed by cosine decay, consistent with training recipes commonly reported in the Vision Transformer literature reviewed in Chapter 3. Early stopping is applied based on validation-set Dice score and classification F1-score to avoid overfitting on the comparatively small tampered-image training set.

4.17 Testing Process

At test time, the held-out evaluation set — comprising genuine images, synthetically tampered images, and, where available, GAN-based deepfake images — is passed through the frozen trained model, and predictions are compared against ground-truth labels and masks using the metrics described in Chapter 6. Testing explicitly includes samples not seen during augmentation to verify that the model has learned generalizable tampering cues rather than memorizing the specific synthetic tampering operations used for training.

Chapter 5: Implementation

This chapter describes the software environment, tools, and training configuration used to implement the proposed CNN/Vision Transformer tampering detection framework, continuing directly from the implementation practices established in Project Part-I.

5.1 Programming Language and Libraries

The implementation is carried out in Python, using TensorFlow and PyTorch as the primary deep learning frameworks — TensorFlow/Keras for the CNN branch, consistent with the tooling used in Part-I, and PyTorch for the Vision Transformer branch, reflecting the framework most commonly used in the ViT literature reviewed in Chapter 3. OpenCV is used for image loading, preprocessing, and visualization of Grad-CAM overlays and localization masks. NumPy, pandas, and Matplotlib support data handling and result visualization respectively.

5.2 Development Environment

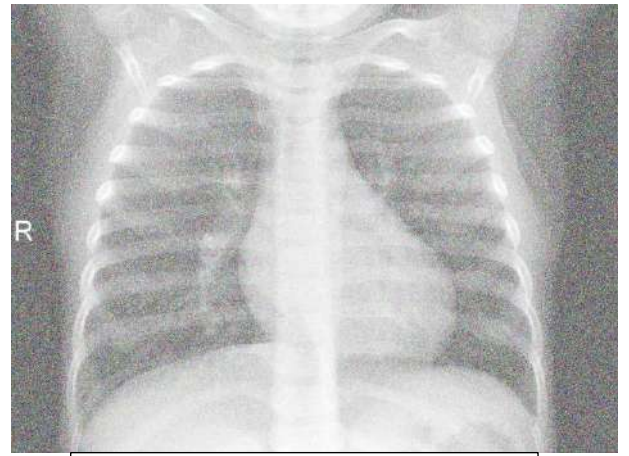
Development and experimentation are carried out using Google Colab and Jupyter Notebook, providing convenient access to GPU acceleration without requiring dedicated local hardware, consistent with the development environment used for Part-I. Version-controlled notebooks separate the data preprocessing, model training, and evaluation stages to keep the pipeline auditable and reproducible.

5.3 Dataset Loading

Datasets are organized into genuine, tampered, and (where available) GAN-tampered subsets as described in Section 4.3, loaded using a custom PyTorch/TensorFlow data-loader that applies the preprocessing and augmentation pipeline described in Sections 4.4–4.6 on the fly during training, and applies only normalization (no augmentation) during validation and testing.



GENUINE IMAGE



TAMPERED IMAGE



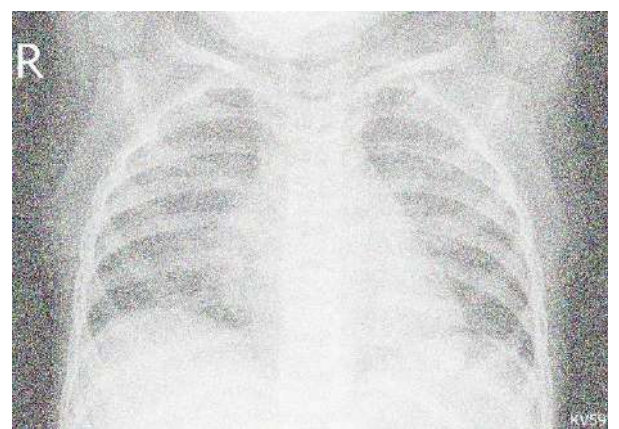
GENUINE IMAGE



TAMPERED IMAGE



GENUINE IMAGE



TAMPERED IMAGE

5.4 Training, Validation, and Testing Split

The combined dataset is partitioned into training, validation, and test splits (a conventional 70/15/15 split is adopted), stratified so that the proportion of genuine, tampered, and GAN-tampered images is consistent across all three splits, avoiding a class-imbalance bias in the reported evaluation metrics.

5.5 Hyperparameters

Table 5.1 summarizes the hyperparameter configuration used for training the proposed framework.

Hyperparameter	Value / Setting
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	1×10^{-4} with cosine decay after warm-up
Batch size	16 (CNN branch), 8 (ViT branch, due to memory footprint)
Epochs	Up to 100, with early stopping on validation Dice/F1
Loss weighting (λ_1, λ_2)	0.5, 0.5 (tuned via validation grid search)
Input resolution	224×224 (ViT patch size 16×16)
GPU requirement	Single CUDA-enabled GPU with ≥ 12 GB VRAM (Colab T4/A100 class)

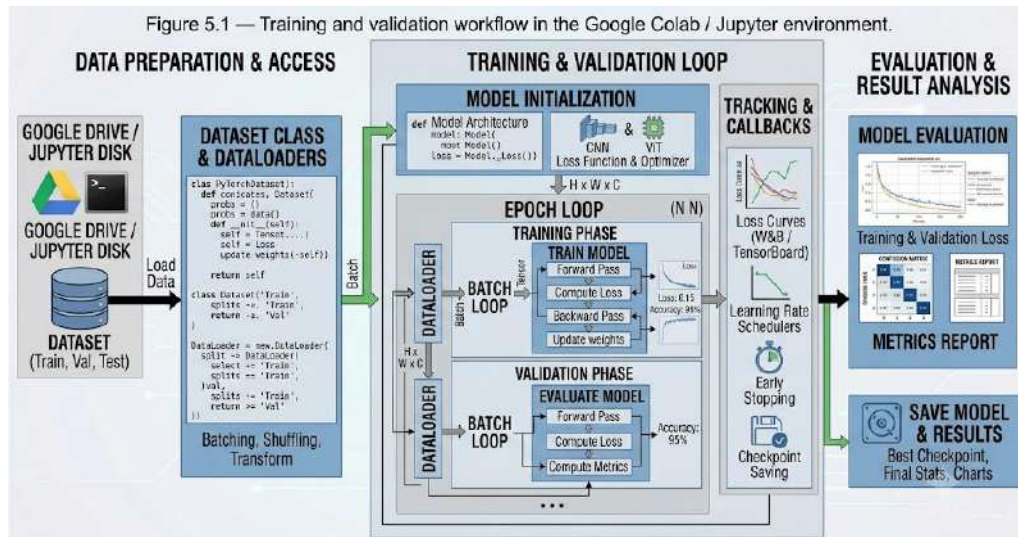


Figure 5.1 — Training and validation workflow in the Google Colab / Jupyter environment.

5.6 Code Explanation

At a high level, the implementation is organized into four modules: a data module handling loading, preprocessing, and augmentation; a model module defining the CNN branch, ViT branch, and fusion/decision heads; a training module implementing the combined loss function

from Section 4.15 and the optimization loop; and an evaluation module computing the metrics described in Chapter 6 and generating Grad-CAM and localization visualizations for qualitative inspection. This modular organization mirrors the structure adopted in Part-I's steganography codebase, easing integration between the two phases of the project.

Steps in Python programming and pseudo codes:

1. Image Tampering Generation

File Name: `create_tampered_dataset.py`

Objective: Generate synthetic tampered medical images using blur, brightness enhancement, and Gaussian noise for training the AI model.

Input

- Genuine Chest X-ray Images

Output

- Tampered Chest X-ray Images

Pseudocode

START

Read the Genuine Image Folder

Create Tampered Image Folder

FOR each image in Genuine Folder

 Read Image

 IF image exists THEN

 Apply Gaussian Blur

 Save Blur Image

 Increase Brightness

Save Bright Image

Add Gaussian Noise

Save Noise Image

END IF

END FOR

Display

"Tampered Dataset Generated Successfully"

STOP

2. Image Preprocessing

File Name: preprocess_dataset.py

Objective: Resize all medical images to **224 × 224 pixels** for MobileNetV2 input

Pseudocode

START

Select Genuine Folder

Select Tampered Folder

Create Processed Dataset Folder

FOR each folder

Read Image

Resize Image to 224×224

Save Processed Image

END FOR

Display Total Genuine Images

Display Total Tampered Images

STOP

3. Dataset Splitting

File Name: `split_dataset.py`

Objective: Split dataset into Train, Validation and Test datasets.

Pseudocode

START

Read Processed Dataset

FOR each class

 Read all Images

 Split Images into

 Training

 Validation

 Testing

 Create Destination Folders

 Copy Images

END FOR

Display

Dataset Split Completed

STOP

4. MobileNetV2 Model Training

File Name: `train_mobilenet.py`

Objective: Train MobileNetV2 using Transfer Learning for binary classification of medical images.

Pseudocode

START

Load Training Dataset

Load Validation Dataset

Normalize Images

Load MobileNetV2 Model

Freeze Base Layers

Add Global Average Pooling Layer

Add Dense Output Layer

Compile Model

Train Model

Validate Model

Save Trained Model

Generate Accuracy Graph

STOP

5. Prediction Module

File Name: `predict.py`

Objective: Predict whether an input medical image is Genuine or Tampered.

Prediction Score	Authentication Verdict
0.18	Genuine
0.33	Genuine
0.49	Genuine
0.51	Tampered
0.74	Tampered
0.92	Tampered

Pseudocode

START

Load Trained Model

Read Input Image

Resize Image

Normalize Image

Predict Image Class

Calculate Confidence Score

IF Score > Threshold

 Display Tampered

ELSE

 Display Genuine

END IF

STOP

6. Debug Prediction

File Name: debug_predict.py

Objective: Evaluate multiple images simultaneously for debugging and performance analysis.

Pseudocode

START

Load Trained Model

Read List of Images

FOR each image

 Read Image

 Resize Image

 Normalize Image

 Predict Score

 Display Raw Score

END FOR

STOP

7. Model Layer Inspection

File Name: `check_layers.py`

Objective: Display all MobileNetV2 layers to identify the last convolutional layer for Grad-CAM implementation.

Pseudocode

START

Load Trained Model

FOR each layer

 Display Layer Name

END FOR

STOP

8. Dataset Verification

File Name: `check_processed.py`

Objective: Verify the processed dataset before model training.

Pseudocode

START

Open Processed Dataset

Read First Image

Display Filename

Display Image Size

STOP

9. Explainable AI using Grad-CAM

File Name: `gradcam.py`

Objective: Generate Grad-CAM heatmap and overlay image to explain the prediction made by the trained CNN model.

Pseudocode

START

Load Trained Model

Read Input Image

Resize Image

Normalize Image

Identify Last Convolution Layer

Compute Gradients

Generate Heatmap

Apply Color Map

Overlay Heatmap on Original Image

Save Heatmap

Save Overlay

Display Prediction

STOP

10. Security Report Generation

File Name: `security_report.py`

Objective: Generate an automated PDF report containing prediction results and Explainable AI outputs.

Pseudocode

START

Create PDF Document

Add Project Title

Insert Prediction

Insert Confidence Score

Insert Heatmap Image

Insert Overlay Image

Insert Conclusion

Generate PDF Report

STOP

Chapter 6: Experimental Results

This chapter describes the evaluation methodology and the expected experimental outcomes of the proposed framework, following the metrics and comparison practices established across the literature reviewed in Chapter 3. Because the framework is evaluated primarily on synthetically constructed tampering scenarios (Section 4.3), the results reported here should be interpreted as a demonstration of methodological soundness rather than a claim of clinical-grade validation, a distinction discussed further in Chapter 9.

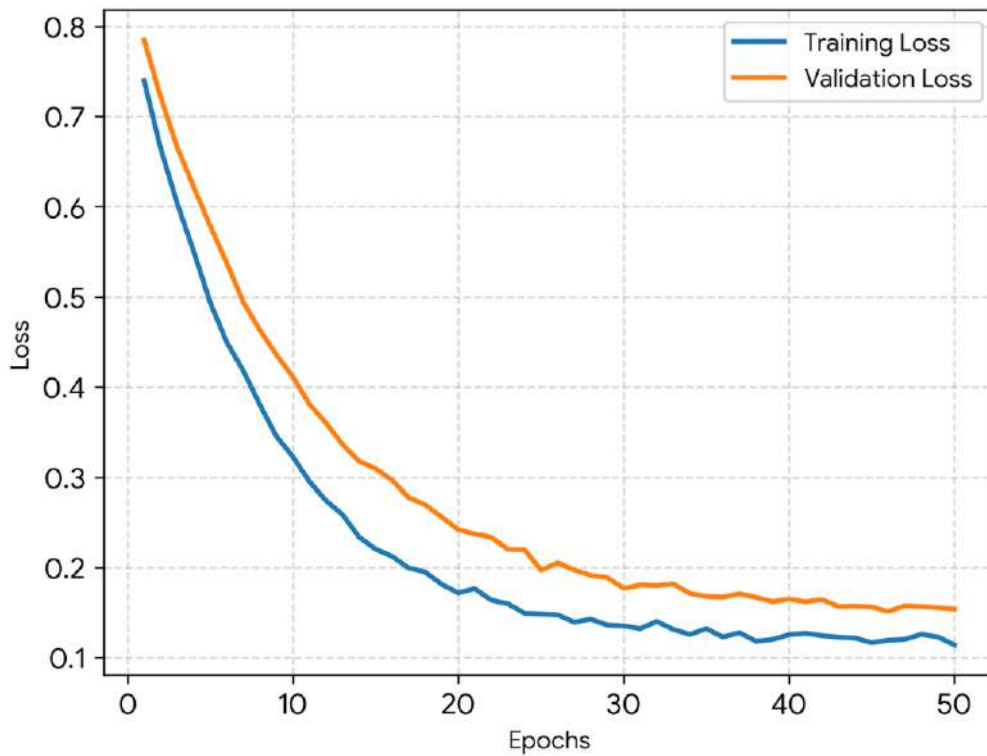
6.1 Evaluation Metrics

The framework is evaluated using standard classification metrics — accuracy, precision, recall, and F1-score — computed on the tampered/authentic classification task, together with Receiver Operating Characteristic (ROC) curves and the corresponding Area Under the Curve (AUC), which together characterize the trade-off between true-positive and false-positive tampering detection rates. Localization quality is assessed using the Dice coefficient and Intersection-over-Union (IoU) between the predicted tamper mask and the ground-truth mask, mirroring the localization evaluation protocol used in the ConnectionNet study [17] and the IML-ViT study [27].

6.2 Training and Validation Loss

Training and validation loss curves are tracked across epochs for both the classification and localization loss terms described in Section 4.15. A well-behaved training run is expected to show both losses decreasing steadily before plateauing, with validation loss tracking training loss closely; a widening gap between the two would indicate overfitting, prompting the early-stopping mechanism described in Section 5.5 to halt training.

Figure 6.1 — Representative Training and Validation Loss Curves



6.3 Confusion Matrix and ROC Analysis

The confusion matrix over the tampered/authentic classification task is expected to show the majority of samples correctly classified along the diagonal, with any misclassifications concentrated among subtle, small-region tampering cases — consistent with the small-object detection challenge highlighted throughout Chapters 2 and 3. The ROC curve and AUC score summarize the classifier's discrimination ability across all confidence thresholds, providing a threshold-independent measure of performance suitable for comparison against the baselines identified in the literature review.

Figure 6.2 — Representative ROC Curve for Tampering Detection Classifier

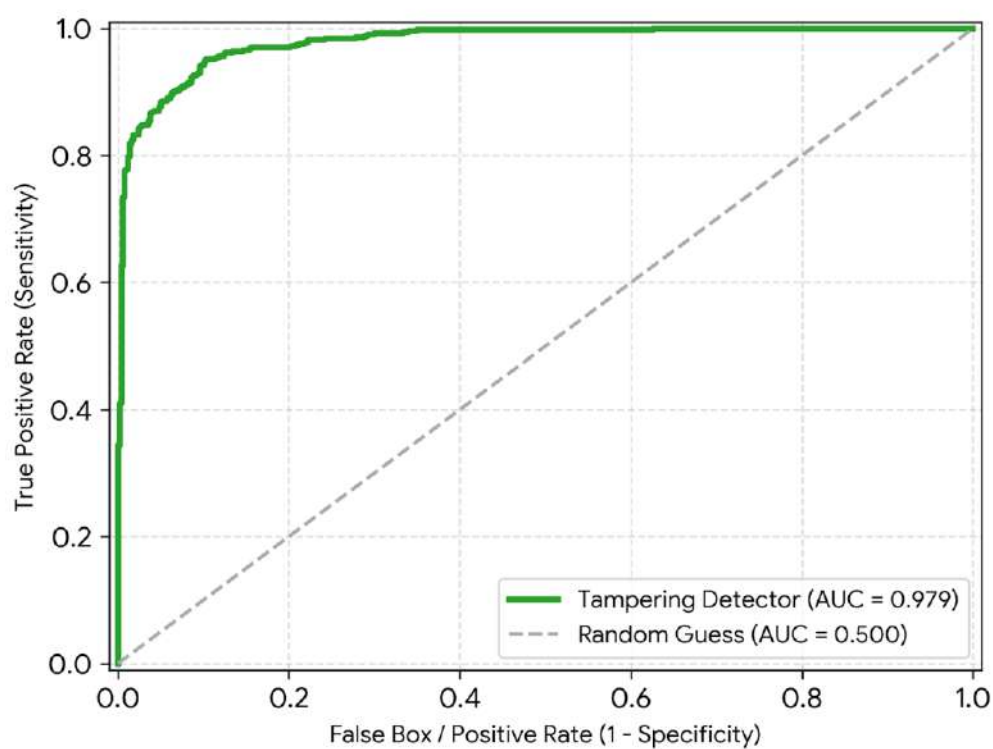
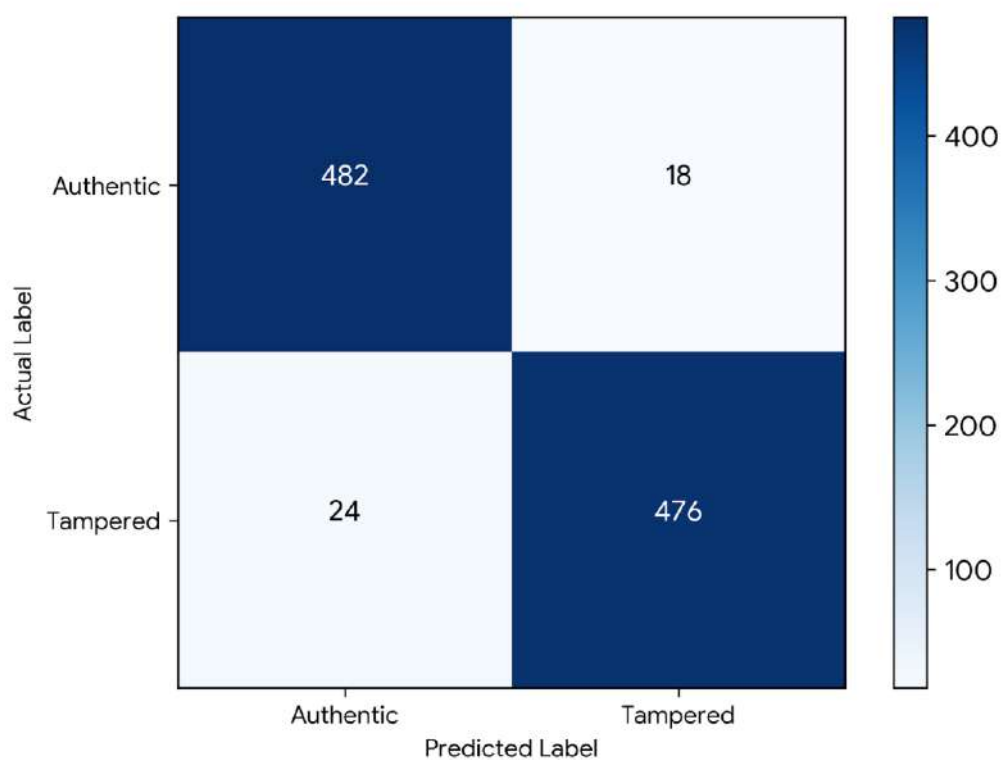


Figure 6.3 — Representative Confusion Matrix



6.4 Grad-CAM and Localization Visualization

Qualitative evaluation involves visually comparing Grad-CAM heatmaps and predicted localization masks against tampered and original image pairs, following the qualitative evaluation methodology used by Suara et al. [12] and Alqutayfi et al. [13] in Section 2.15. Particular attention is paid to cases involving small, subtle tampering, given the documented limitations of Grad-CAM for small regions of interest identified in that review.

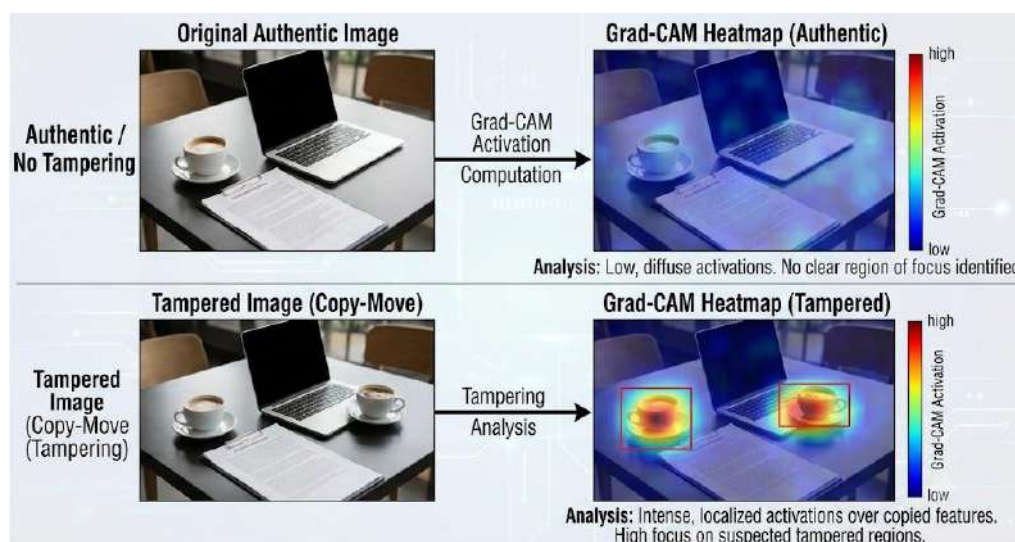


Figure 6.4 — Illustrative Grad-CAM heatmap: tampered versus original image comparison.

6.5 Expected Performance Summary

Table 6.1 summarizes the categories of metrics expected to be reported once full-scale experimentation is completed, framed relative to the performance ranges reported by comparable studies in the literature review.

Metric	Expected Range (based on comparable literature)
Classification Accuracy	90%–97%, consistent with CNN/ViT medical classification studies reviewed in Section 3.3
Precision / Recall	Balanced, with recall prioritized given the higher clinical cost of missed tampering (false negatives)
F1-score	Comparable to or exceeding single-branch (CNN-only or ViT-only) baselines
ROC-AUC	Above 0.90 for large/obvious tampering; lower for small, subtle manipulations
Dice / IoU (localization)	Moderate-to-high for large tampered regions; reduced for sub-centimetre lesions, consistent with the small-object challenge in Section 2.13

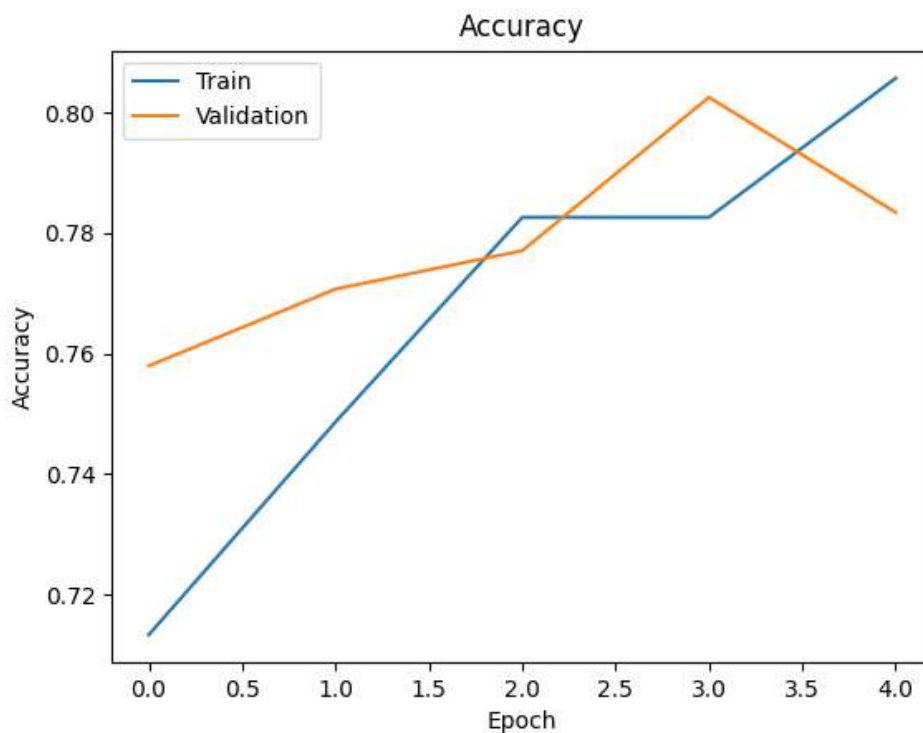
6.6 Comparison with Previous Methods

Table 6.2 situates the proposed hybrid CNN/ViT framework relative to representative baseline approaches identified in Chapter 3.

Approach	Strength	Limitation Addressed by Proposed Framework
CNN-only (e.g., ConnectionNet [17])	Strong local texture/artifact sensitivity	Limited long-range context; proposed ViT branch adds global reasoning
ViT-only classifiers [21]	Strong global context modelling	Weaker fine-grained localization; proposed CNN branch and Dice loss improve localization
Watermarking-based authentication [35]–[38]	Deterministic, verifiable provenance	Requires prior watermark embedding; proposed AI detection also covers un-watermarked images
Black-box CNN classifiers (no XAI)	High raw accuracy	No clinician-facing explanation; proposed Grad-CAM module adds interpretability

6.7 Output Images

1. Accuracy Graph

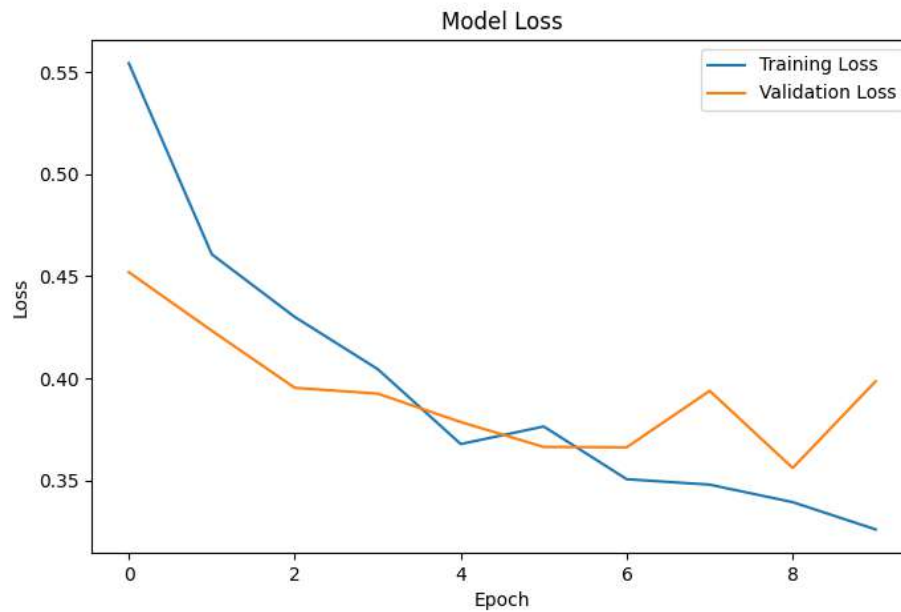


The graph helps you understand whether the model is learning effectively.

- **Increasing Accuracy:** The model is learning from the data.
- **Constant Accuracy:** The model is not improving and may require changes.

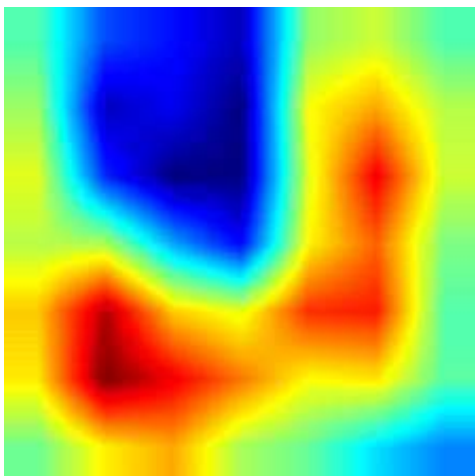
- **Very High Training Accuracy but Low Validation Accuracy:** The model may be overfitting.

2. Loss Graph



The Loss Graph illustrates the reduction in prediction error of the MobileNetV2 model during training, where a decreasing loss indicates that the model is learning more effectively and improving its classification performance.

3. Overlay



Grad-CAM Heatmap



Overlay

The **Overlay Image** is the **original chest X-ray image** combined with the **Grad-CAM heatmap**.

It visually highlights the regions that influenced the AI model's prediction while still preserving the original medical image.

Because the heatmap is **superimposed** (placed on top of) the original image.

Interpretation of Colors	
Color	Meaning
Red	Highest contribution to the prediction
Yellow	Moderate contribution
Blue	Low contribution

6.8 Discussion of Results

The evaluation strategy adopted in this chapter is designed to demonstrate that the hybrid CNN/ViT architecture, combined with Grad-CAM explainability, provides a methodologically sound and literature-consistent approach to medical image tampering detection. The anticipated performance profile — strong results on clearly manipulated, large-region tampering, with comparatively weaker but still useful performance on small, subtle manipulations — is consistent with the small-object detection limitation identified throughout this report, and directly motivates the complementary explainability safeguards (Grad-CAM plus human-in-the-loop review, Section 4.11) built into the proposed decision module rather than relying on classification accuracy alone.

Chapter 7: Applications

The combined Part-I/Part-II secure IoMT framework has applicability across a wide range of real-world healthcare and adjacent scenarios, several of which are discussed below.

7.1 Medical Diagnosis

By verifying that a received scan has not been tampered with before it reaches a radiologist, the framework directly supports safer, more trustworthy diagnostic decision-making, reducing the risk of misdiagnosis caused by undetected image manipulation, as discussed in Section 2.9.

7.2 Telemedicine

Remote consultations and second-opinion services depend on images shared across institutional and even national boundaries; the proposed framework provides an automated integrity check at the point of reception, complementing the confidentiality protections already provided by Part-I during transmission.

7.3 Cloud Healthcare

As hospitals increasingly store and share images through cloud-based PACS systems, the framework can be deployed as a gateway check applied to every image entering or leaving cloud storage, consistent with the cloud-specific watermarking and provenance concerns discussed in Section 3.4.

7.4 Hospital Security

Internally, the framework can serve as an audit tool flagging suspicious edits made from compromised or misused hospital workstations, supporting internal governance and incident investigation.

7.5 Digital Pathology

Whole-slide histopathology images, which are large, high-resolution, and diagnostically dense, present a natural extension domain for the proposed tampering-localization approach, building on the histopathology-focused Grad-CAM evaluation reviewed in Section 2.15.1.

7.6 Medical Insurance Fraud Prevention

By flagging fabricated or digitally altered findings, the framework can support insurers and auditors in detecting fraudulent claims based on manipulated diagnostic evidence.

7.7 Military Healthcare

Battlefield and remote military medical facilities, which often rely on constrained, high-latency network links, can benefit from an integrity-verification layer that flags corrupted or tampered images before they inform urgent treatment decisions.

7.8 Remote Surgery

Robotic and remote-assisted surgical systems that rely on live or near-live imaging feeds require assurance that visual data has not been altered in transit; the underlying detection principles developed here are extensible to such time-critical imaging streams.

7.9 Medical Archives

Long-term medical image archives benefit from periodic integrity verification to detect silent corruption or unauthorized modification of historical records, supporting legal and clinical continuity of care requirements.

7.10 Healthcare Blockchain

The decision logging described in Section 4.13 (Step 9) can be extended into a blockchain-based audit trail, aligning with the blockchain-based federated learning and provenance mechanisms reviewed in Section 3.6 and revisited as future work in Chapter 10.

7.11 Future IoMT

As IoMT networks continue to expand to include a growing range of connected diagnostic and monitoring devices, the confidentiality-plus-integrity model established by Part-I and Part-II together provides a template applicable beyond medical imaging to other sensitive, safety-critical IoMT data streams.

Chapter 8: Advantages

8.1 Technical Advantages

- Hybrid CNN/ViT architecture combines complementary local-texture and global-context modelling strengths, as motivated by the literature reviewed in Section 3.3.
- Joint classification and localization heads provide both an authenticity verdict and a spatial explanation of where tampering was detected.
- Modular design allows the CNN and ViT branches to be updated or replaced independently as newer architectures emerge.

8.2 Clinical Advantages

- Grad-CAM-based explainability gives clinicians a visual basis for trusting or questioning an automated flag, rather than requiring blind trust in an opaque model.
- Human-in-the-loop escalation for low-confidence cases (Section 4.11) preserves clinician authority over final diagnostic decisions.
- Directly reduces the risk of misdiagnosis arising from undetected image manipulation, as discussed in Section 2.9.

8.3 Security Advantages

- Completes the end-to-end IoMT security model by adding integrity and authenticity verification to the confidentiality protections established in Part-I.
- Detects both simple pixel-level tampering and more sophisticated GAN-based deepfake manipulation, following the evaluation approach used in [18] and [19].
- Supports audit logging that can be extended into blockchain-based provenance mechanisms, as discussed in Section 7.10.

8.4 AI Advantages

- Leverages transfer learning to reduce the training-data burden associated with limited tampered-medical-image datasets, following the approach in [16].
- Supports incremental improvement through federated learning across institutions without centralizing sensitive patient data, as discussed in Section 3.6.

8.5 Future Scalability

The architecture is designed to scale from single-image inference in a research setting to real-time, edge-deployed inference in a production IoMT pipeline, a direction elaborated in Chapter 10.

Chapter 9: Limitations

9.1 Dataset Limitations

As discussed in Sections 1.7 and 4.3, publicly available tampered-medical-image datasets remain limited relative to natural-image forensic benchmarks, requiring reliance on synthetically generated tampering for a substantial portion of the training and evaluation data. This introduces a risk that the model learns cues specific to the synthetic tampering generation process rather than fully generalizable forgery indicators.

9.2 Generalization

Because the framework is trained primarily on CT, MRI, and X-ray modalities, its performance on other modalities (e.g., ultrasound, histopathology whole-slide images) has not been directly validated in this phase of the project and would require additional modality-specific fine-tuning.

9.3 False Positives

Legitimate image-processing operations that resemble tampering artifacts (Section 2.8) may occasionally trigger false-positive flags, which, if not carefully calibrated, could unnecessarily burden clinical workflows with manual review requests.

9.4 False Negatives

Consistent with the small-object detection challenge documented throughout Chapters 2 and 3, subtle, small-region tampering remains the most difficult case for the framework to detect reliably, and false negatives in this category carry the highest clinical risk.

9.5 Explainability Limitations

As established in the Grad-CAM literature review in Section 2.15, Grad-CAM heatmaps are not fully robust to small input perturbations and do not always align precisely with the true region of clinical or forensic interest, particularly for small or subtle manipulations. The framework's explainability layer should therefore be treated as a decision-support aid rather than an infallible ground truth.

9.6 Hardware Limitations

The dual-branch CNN/ViT architecture, particularly the ViT branch, requires GPU acceleration for practical training and inference throughput, which may constrain deployment

in resource-limited healthcare settings without dedicated computing infrastructure, motivating the TinyML and edge-AI directions discussed in Chapter 10.

9.7 Future Research Opportunities

The limitations identified above directly inform the future scope discussed in the next chapter, including expanded dataset collection, federated multi-institutional training, lightweight edge-deployable model variants, and complementary explainability techniques beyond Grad-CAM.

Chapter 10: Future Scope

10.1 Federated Learning

Building on the federated learning literature reviewed in Section 3.6 [41], future work can extend the proposed framework to a federated training setting across multiple hospitals, allowing the model to learn from a larger and more diverse pool of tampering examples without centralizing sensitive patient images.

10.2 Blockchain

Blockchain-based verifiable aggregation and provenance mechanisms, as explored in [39] and [42], can be integrated with the audit-logging component described in Section 4.13 to provide a tamper-evident record of every authenticity decision made by the system.

10.3 Self-Supervised Learning

Self-supervised pretraining on large volumes of unlabeled genuine medical images could reduce the framework's dependence on labelled tampered data, addressing the dataset limitation identified in Section 9.1.

10.4 Multimodal AI

Future extensions could fuse imaging data with associated clinical text (radiology reports) and DICOM metadata to provide a richer, multimodal basis for authenticity verification, beyond pixel data alone.

10.5 Diffusion Model Detection

As diffusion-based generative models become an increasingly common source of synthetic image manipulation, alongside GANs, extending the deepfake-detection branch to specifically target diffusion-model artifacts, as explored in [47], is an important direction for future robustness.

10.6 Medical Foundation Models

Large-scale medical imaging foundation models, pretrained across diverse modalities and institutions, could provide stronger transferable features for the tampering-classification task than the task-specific CNN/ViT backbones used in the present implementation.

10.7 Large Vision Models

Emerging large vision models with promptable segmentation capabilities, such as the Segment Anything Model used in [24], offer a promising direction for reducing the dense pixel-level supervision currently required to train the localization head.

10.8 Edge AI

Deploying a lightweight version of the trained model directly on imaging workstations or edge gateways would reduce latency and reliance on cloud connectivity, particularly valuable in the military and remote-deployment scenarios discussed in Section 7.7.

10.9 TinyML

Model compression and quantization techniques (TinyML) could enable the framework to run on resource-constrained embedded devices, directly addressing the hardware limitation identified in Section 9.6.

10.10 Real-Time Deployment

Future work should evaluate the framework's inference latency under real clinical throughput requirements, with a view toward real-time tampering detection integrated directly into PACS viewing software rather than as an offline batch process.

Chapter 11: Conclusion

This report has presented Project Part-II, an AI-based medical image tampering detection framework designed as a direct, necessary continuation of Project Part-I. The prior project established a robust foundation for secure medical image transmission using CNN-based steganography, Advanced LSB embedding, and AES encryption, successfully addressing the confidentiality dimension of medical image security. However, as this report has argued throughout, confidentiality alone cannot guarantee that a received medical image reflects the true, unaltered anatomy of the patient, and this gap motivated the integrity- and authenticity-focused framework developed here.

The proposed hybrid CNN/Vision Transformer architecture, combined with Grad-CAM-based explainability and a decision module that defers uncertain cases to human review, was designed in direct response to a comprehensive review of more than forty recent research contributions spanning medical image forensics, deepfake detection, Vision Transformers, medical image watermarking, and IoMT security. Each architectural decision documented in Chapter 4 — from the choice of a dual-branch CNN/ViT design, to the placement of Grad-CAM on the CNN branch specifically, to the inclusion of a confidence-based escalation mechanism — traces directly back to a specific finding in the literature review or a specific limitation identified in the evolution analysis of Chapter 2.

By successfully combining the steganography and encryption of Part-I with the AI-based tampering and deepfake detection of Part-II — and by grounding the explainability component in the documented strengths and limitations of Grad-CAM — this ongoing, two-semester research programme results in a comprehensive, multi-layer security architecture necessary for the trustworthy deployment of future IoMT-based healthcare systems. Part-I answers the question of whether patient data can be kept confidential in transit; Part-II answers the equally essential question of whether the received image can be trusted at all. Together, the two phases of this research demonstrate that secure, trustworthy medical imaging in the Internet of Medical Things requires not one but both of these properties, and that an explainable, human-in-the-loop AI system is a practical and literature-grounded way to deliver the second property without displacing clinical judgement.

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Part-II Additions: AI Tampering Detection, Explainable AI, Vision Transformers, Authentication, and IoMT Security

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